

AMetric Normal-Form Homology

Origin-Side Licensing, Endpoint-Side Exhaustion, and the Black-Hole Residue Application
under AASC

AASC Black-Hole Endpoint Arc, Paper III: AMetric Boundary Exposure and Homology

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Abstract

This paper proves an AASC-internal theorem of AMetric normal-form homology. The dependency order is fixed and one-way: the AASC kernel forces the admissibility-bearing roles of admissibility, standing, reference, and irreversibility; the AMetric invariant fixes the boundary normal form; standing-bearing physical interiors, metricity, Einstein dynamics, initial data, horizons, and black-hole reconstruction are licensed only downstream. No metric-interior object, equation, singularity, horizon, or endpoint authorizes the kernel or the AMetric invariant.

AASC is used here as a constraint formalism. Its operational work is boundary exhaustion and impossibility, not the proposal of a successor cosmology, endpoint mechanism, or preferred continuation law. Once the kernel, AMetric invariant, origin-side licensing profile, and endpoint-side exhaustion target are fixed, every standing-relevant same-scope alternative must appear in the declared role-profile coordinates, resolve through a certified exterior witness, enter independently governed richer scope, or fail under route closure.

At the origin side, the AMetric profile excludes primitive metric, temporal, causal, probabilistic, selector, operator, and inherited-law authority. What ordinary physics calls an initial condition therefore enters only after admission, as a declared anchor, quotient-level tensor, or transition certificate inside a licensed continuation package. Cauchy data, low-entropy cosmological constraints, scalar-field starting values, vacuum choices, and Past-Hypothesis-style conditions may be physically indispensable without becoming primitive sources of their own standing.

The admitted interior may then support a faithful four-dimensional Lorentzian projection of the AASC gravity role. In the declared minimal first-response same-scope class, its governed metric/source equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

This Einstein projection is licensed and bounded. It provides the continuation envelope in which ordinary relativistic initial-value practice, source closure, clocks, records, and causal description have standing; it does not acquire authority over the boundary conditions that licensed it.

The endpoint profile is derived clause by clause from downstream role-support exhaustion. When the old-scope metric, temporal, causal, source-ledger, and operator roles no longer have standing-bearing support, and no independent re-governance, repair, selector, transport, storage, dynamical-transition, inherited-law, or admitted-null route remains, the endpoint has the independently fixed AMetric boundary role profile. The black-hole residue theorem supplies the target-class exact physical instantiation. For a certified AASC black-hole target, once exterior-quotient and exterior-witness subtraction leave a nonempty unresolved horizon-isolated residue, old-scope standing transport is undefined, residual metric successor support is empty, no independent re-governance precedes exhaustion, and admitted nullity is closed, the endpoint is the New AMetric Boundary. Hair-like, dynamical, or richer metric structure avoids endpoint activation only to the extent that it genuinely re-fixes standing before the residue is formed or declares a new governance scope before exhaustion; no third old-scope terminal profile remains.

The homology capstone result is

$$B_A^\star \neq B_A^0, \quad \mathcal{P}_{\text{ANF}}(B_A^\star) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}, \quad B_A^\star \cong_{\text{ANF}} B_A^0.$$

The profiles are token-distinct and role-profile identical. Their homology is structural rather than temporal, and classificatory rather than transmissive. The black-hole bivalent-resolution theorem follows as a downstream corollary: each declared horizon-generated burden is either re-fixed by a standing-bearing exterior witness relation or remains unresolved; no third old-scope standing status exists. None of these results implies cyclic cosmology, a crossing map, inherited information or energy, common law, causal production, or a populated successor domain.

Exact theorem endpoint

The AASC kernel and AMetric invariant are dependency-prior to every physical interior. They classify the origin-side boundary at which standing-bearing physical description is licensed and the endpoint-side boundary exposed when a downstream governed regime exhausts its old-scope role support. The theorem establishes token distinction together with equality of the complete AMetric boundary role profiles. Black-hole residual closure supplies the certified endpoint instantiation for the declared target class; it does not authorize the invariant. For that certified target, the New AMetric Boundary is forced rather than selected. No crossing, inheritance, cyclicity, causal production, or populated successor follows.

Kernel-deletion firewall

The homology theorem presupposes the full AASC kernel. Removing admissibility erases the difference between endpoint route closure and arbitrary redescription: a singularity label, final-state rule, remnant index, tunneling slogan, hidden branch, or metric continuation could then be treated as a terminal status without passing a route test. Removing standing erases the evaluated standing-bearing residue: the endpoint would no longer be a failure of old-scope support for a threatened burden, but only an untyped change of description. Removing reference erases the target whose support exhausts: R_H , $B_A^\star$, and the old-scope lineage would no longer have determinate roles. Removing irreversibility permits the endpoint to be repaired after failure and then counted as the same lineage.

Thus kernel denial is not an alternative endpoint profile. It destroys the origin-endpoint comparison object. Endpoint exposure is meaningful only because Adm sorts endpoint routes, Stand supplies the standing-bearing burden, Ref fixes the residue being classified, and Irr prevents a false endpoint from repairing an exhausted route.

AASC Black-Hole Endpoint Arc Position

Shared arc spine: kernel/AMetric invariant $\rightarrow$ licensed AASC-GR black-hole regime $\rightarrow$ exterior no-totalization $\rightarrow$ standingless residual interior $\rightarrow$ governance-depth exhaustion $\rightarrow B_A^\star \rightarrow$ AMetric homology $\rightarrow$ fresh internal governed domain.

Shared series abstract. The AASC Black-Hole Endpoint Arc is a five-manuscript sequence that treats black holes as a fixed-domain stress test for AASC. It first role-types the licensed black-hole regime, then proves the exterior no-totalization obstruction, then classifies the uncertified horizon-isolated carrier as residual coherence rather than ordinary standing-bearing interiority, then exposes the endpoint AMetric normal form and no-crossing homology, and finally classifies the counter-boundary survivor as fresh internal governed-domain status. The sequence is conditional and one-way: later endpoint or capstone conclusions never authorize earlier target formation.

Paper label. Paper III / AMetric boundary exposure and homology. This manuscript exposes the endpoint AMetric normal form and proves token-distinct origin-endpoint role-profile equality with no crossing.

This manuscript is Paper III of the arc. Its job is boundary exposure and homology, not universe production. It compares the origin-side AMetric licensing profile with the endpoint-side boundary profile exposed after downstream role-support exhaustion.

Imported into this paper. The paper imports the exterior no-totalization result and the standingless-interior/residual-coherence bridge: unresolved black-hole residue has no old-scope standing transport, the horizon is not an AMetric reset, residual organization is non-standing, and maximum governance-depth is reached when residual metric successor support is exhausted.

Proved here. The paper proves endpoint-side AMetric exposure and normal-form homology. Under the certified target conditions, the endpoint token $B_A^\star$ is distinct from the origin token B_A^0 while sharing the same complete AMetric role profile: $\text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \mathbf{aANF}$. The relation is structural and classificatory, not temporal, causal, genealogical, or transmissive.

Exported to the next paper. The export is the handoff package for the capstone: token-distinct AMetric homology, absolute no-crossing, no inheritance, null-route closure, and the refusal of cyclicity, causal production, energy transfer, information transfer, or populated successorhood at the homology stage.

Not claimed here. This paper does not prove that black holes make universes, does not select a successor cosmology, and does not infer a fresh internal governed domain. It stops at homology plus no-crossing; the counter-boundary occupation result is a later theorem.

Series dependency lock. This paper begins only after the exterior obstruction and residual-coherence bridge have supplied the certified endpoint antecedents. It does not import the counter-boundary occupation theorem. In particular, the implication used here is the endpoint-side one

$$R_H \neq \emptyset, \quad \text{Succ}_{D_0}^{\text{res}}(R_H, \text{Metric}) = \emptyset, \quad \text{closed endpoint routes} \implies \text{Status}(R_H) = B_A^\star.$$

The comparison $B_A^\star \cong_{\text{ANF}} B_A^0$ is a role-profile comparison only. It supplies no crossing map, successor occupation, or fresh-domain existence claim.

Forward handoff. The capstone may import only the completed homology handoff: token distinction, equal AMetric normal-form profiles, absolute no-crossing, no inheritance, null-route closure, and the fact that homology alone yields no populated or causal successor. The capstone must add its own counter-boundary route exhaustion before asserting fresh internal governed-domain status.

Referee-facing arc firewall. Normal-form homology is the most vulnerable point for overreading, so the theorem is locked to role-profile comparison. The equality $\text{PANF}(B_A^\star) = \text{PANF}(B_A^0) = \mathbf{aANF}$ does not provide a path, bridge, time order, causal production relation, common Hilbert space, conserved cross-boundary ledger, inherited law package, or populated successor. Any proposal that supplies one of those structures has either declared a richer third-domain scope or denied the no-crossing target rather than extracted it from homology.

This firewall is also the anti-circularity guard for the capstone. The homology theorem may export token distinction, endpoint normal-form equality, no-crossing, no inheritance, and null-route closure. It may not export counter-boundary occupation. A same-scope objection to this paper must therefore exhibit a missing or misclassified normal-form coordinate, a surviving same-scope endpoint route, a break in the endpoint antecedents, or a misuse of an imported AASC result. Asking homology to produce a universe is a request for the next paper, not an objection to this one.

Pass 6 notation lock. In this manuscript R_H denotes the unresolved horizon-isolated residue after exterior-witness subtraction; it is not the exterior reference family R_{ext} . The tokens B_A^0 and B_A^* are boundary-position tokens, not spacetime locations, metric surfaces, or linked events. The map P_{ANF} or PANF assigns a boundary token its AMetric role-profile vector, and a_{ANF} is the complete normal-form vector. Thus $\text{PANF}(B_A^*) = \text{PANF}(B_A^0) = a_{\text{ANF}}$ states role-profile equality only. The capstone symbols D_1 , FreshGov_{CB} , and $\equiv_{\text{gov},CB}$ are reserved for Paper IV and are not conclusions of the homology theorem.

Certificate terminology lock. The word *certificate* has four arc-level uses, and the uses are not interchangeable. A *standing-transport certificate* is a same-scope map or witness assignment that preserves a threatened reference class in the exterior reconstruction problem. An *exterior witness certificate* is a concrete exterior, boundary, radiation, remnant, holographic, island, or quotient map that removes a burden from the unresolved-residue ledger. An *endpoint-certified target* is not a rescue certificate; it is the antecedent package in which a nonempty unresolved residue, old-scope standing loss, residual metric-support exhaustion, closed false endpoints, null-route closure, and no prior re-governance make endpoint AMetric exposure the target. A *capstone-certified endpoint* is the Paper IV input: the homology handoff together with endpoint certification and no-crossing. A horizon can be transport-certified and thereby exit the exterior obstruction; an unresolved residue can be endpoint-certified precisely because no such transport certificate has resolved it.

Exterior-quotient and metric-regime lock. The proof-level object of the arc is the declared exterior standing quotient of the black-hole model, not a particular metric-regime representative. In the familiar settled Kerr–Newman example the quotient is displayed by mass, electric charge, and angular momentum, but that example does not restrict the theorem to stationary representatives and does not make absence of hair a separate proof premise. Hair-like, dynamical, evaporating, ringdown, perturbative, semiclassical, quasi-local, high-curvature, or quantum-gravity structure has same-scope force only when it is declared as exterior quotient content, an exterior witness, a leakage channel, a certificate, a non-sink quotient, or richer governance. Such structure subtracts from the unresolved residue exactly to the extent that it re-fixes standing-bearing content. If it does not re-fix standing, it is skin, bookkeeping, hidden load, repair, selector import, carrier laundering, or scope change; it does not provide an additional endpoint route.

Endpoint exposure after exterior-quotient subtraction. Paper III begins endpoint exposure only after all declared exterior quotient content and all certificate-bearing structures have performed their subtractive work. Classical mass, charge, and angular momentum parameters, additional hair-like witnesses, radiation correlations, perturbative records, boundary maps, holographic or island certificates, remnant maps, non-sink quotients, or richer-scope structures remove exactly the burdens they re-fix. The endpoint proof begins with the nonempty unresolved remainder R_H . It does not assume stationarity, absence-of-hair metaphysics, or absence of every possible hair-like structure. It assumes only that the declared quotient and certificates leave a nonempty unresolved residue whose old-scope standing transport is undefined and whose residual metric-support exhausts.

Endpoint exposure self-containment lock. For the five-paper arc, this manuscript does not rely on a missing successor-cosmology theorem. From Paper II it imports only the standingless residual-coherence classification, the no-horizon-as-AMetric-reset result, and the condition that residual metric-comparison successor support may exhaust. Given those exports together with the AASC kernel, AMetric boundary discipline, null-route closure, no-selector/no-repair discipline, and fixed-domain route exhaustion, this manuscript locally classifies the endpoint role profile. The token

B_A^* names the endpoint-side AMetric boundary only after those coordinates have been independently populated. It is not a counter-boundary occupant, not a transported successor, and not an import from Paper IV.

Pass 9 local referee-response lock. For local review, the endpoint step is not a hidden appeal to the counter-boundary capstone. The local route is: Paper II supplies standing break, no horizon-as-AMetric-reset, residual coherence, and the meaning of residual metric-support exhaustion; this paper then performs the endpoint-coordinate population and false-endpoint/null-route closure that classify the certified unresolved target as B_A^* . The proof load is carried locally by the endpoint-coordinate population, endpoint-side boundary exposure, and black-hole AMetric endpoint exactness clauses, after which normal-form homology compares B_A^* with B_A^0 . A same-scope referee objection must therefore challenge one of those endpoint coordinates or route closures. It may not treat homology as algebraic-topological homology, a bridge, a transition, cyclicity, common geometry, or fresh-domain existence.

Pass 10 hostile-referee closure lock. The local self-containment test is this: Paper II exports the residual-support antecedents, but this paper performs the endpoint classification by populating the AMetric profile coordinates and closing the false endpoint, null, repair, selector, metric, temporal, causal, storage, and inherited-law routes. A critic may deny that the certified black-hole target reaches residual metric-support exhaustion; that denies the endpoint target. A critic may exhibit an unclosed endpoint coordinate; that would challenge this paper. But a request for a post-boundary domain, transition mechanism, baby-universe channel, common Hilbert space, or fresh governance object belongs to Paper IV or a richer-scope theory. Homology here remains only token-distinct role-profile equality plus no crossing and no inheritance.

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Roadmap and exact claim

The proof proceeds from primitive constraint to downstream application.

1. The AASC kernel fixes admissibility, standing, reference, irreversibility, target identity, and fail-closed reuse (Maley, 2026g).
2. The admissibility-structure and AMetric theorems fix the regime-invariant boundary normal form: no metric, order, probability, selector, canonical representative, operator, repair, or cross-domain standing transfer has boundary authority; standing is bivalent, reuse-stable, and unavailable to same-scope creation or repair (Maley, 2026i,n,p).
3. The origin-side application shows that initial conditions are admitted only as anchors, tensors, or certificates after boundary admission; standing monotonicity prevents their reinterpretation as primitive temporal beginnings (Maley, 2026b,e,j).
4. The realized-interior result licenses the physical continuation package downstream of the invariant (Maley, 2026l). Metricity is licensed within that package (Maley, 2026f). The gravity-role result fixes the constraint architecture (Maley, 2026d). The Einstein projection fixes the declared metric/source form (Maley, 2026c).
5. Endpoint-side exposure is tested against the already fixed normal form. The residual-coherence and black-hole endpoint theorems supply the certified target in which a licensed Einsteinian regime loses residual metric authority for a nonempty unresolved residue; at maximum governance-depth the New AMetric Boundary is forced (Maley, 2026h,k,m).
6. Independent derivation of the origin-side and endpoint-side role-profile coordinates proves AMetric normal-form homology. The bivalent black-hole burden theorem follows as an application corollary, not as proof authority for the invariant.

Invariant first, application second

The theorem does not infer AMetricity from black-hole physics. The black-hole case is admissible only downstream of the kernel and AMetric invariant. At endpoint, the invariant classifies the exhaustion profile of that downstream regime. Reversing the order is boundary backflow.

The theorem object

The homology capstone theorem is the equality of the complete AMetric boundary role profiles at two token-distinct theorem positions. It is not the kernel theorem, the AMetric invariant, the Einstein projection, the black-hole endpoint theorem, or the bivalent burden partition taken separately. Its object is the dependency-ordered relation between origin-side licensing and endpoint-side old-scope exhaustion.

AASC constraint-formalism method

AASC is a constraint formalism. Its operational mode here is boundary exhaustion and impossibility rather than mechanism selection. The contribution is closure over a fixed origin-endpoint comparison class, not a black-hole production model, bounce dynamics, successor cosmology, or preferred endpoint law. Once the target is fixed, a standing-relevant alternative must change an evaluated boundary coordinate, alter retained continuation, resolve the declared burden through a certified witness, introduce independently governed richer scope, or fail. No untyped same-scope remainder is available.

The nontrivial work is the independent coordinate derivation. The origin profile is fixed by pre-admission closure. The endpoint profile is fixed only after a standing-bearing interior, a licensed metric/source envelope, residue formation, exterior-witness subtraction, residual-support exhaustion, null-route closure, and exclusion of inherited continuation. Equality is concluded only after both vectors have been populated independently.

Established background. General relativity supplies metric-internal horizon, singularity, causal, source-ledger, and initial-value descriptions. Quantum-gravity programs may supply exterior reconstruction, holographic maps, larger Hilbert spaces, tunneling amplitudes, baby-universe sectors, bounce dynamics, or independently governed post-boundary domains.

Constraint-formalism result. The theorem does not rank those mechanisms. It fixes the general AASC boundary comparison class and proves that origin-side licensing and endpoint-side old-scope exhaustion have the same complete AMetric role profile. The certified black-hole package then instantiates the endpoint exactly: an unresolved residue at maximum governance-depth has New AMetric Boundary status, with no third old-scope terminal profile.

Theorem boundary. The result does not infer endpoint status from a mere astronomical label, a metric representative, or a stationary or absence-of-hair slogan. The relevant black-hole target is the post-subtraction object: the declared exterior quotient and every certificate-bearing witness have removed exactly what they re-fix, leaving a nonempty unresolved residue whose residual metric support exhausts. A model that resolves the residue, declares richer governance before exhaustion, or claims only the coarser exterior quotient is not a counterexample; it is a different target disposition. A model that preserves the unresolved burden while seeking a same-scope terminal survivor must meet the route-closure burden below.

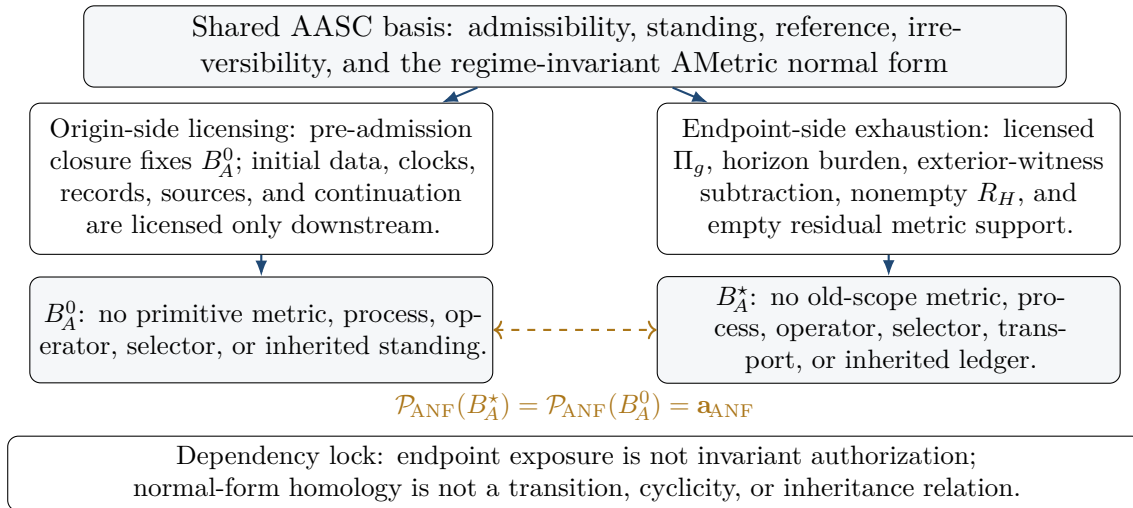


Figure 1: Constraint-formalism architecture of AMetric normal-form homology. The origin and endpoint branches share an upstream invariant but no physical crossing map.

Standing of objections

A criticism has standing only if it preserves the fixed homology target and kernel-faithful comparison class and then denies target certification, breaks a load-bearing imported theorem,

identifies an invalid specialization or dependency use, proves that the capstone antecedents cannot jointly instantiate, or exhibits a faithful same-scope survivor outside the role-profile and route-exhaustion normal forms. Requests for a familiar mechanism, a physical crossing story, a cyclic interpretation, a preferred successor, a weaker selector, or a populated-domain theorem do not challenge this result unless they supply one of those witnesses.

Correctness burden

The correctness routes are exact:

deny the target $\vee$ break an import $\vee$ show dependency misuse
 $\vee$ prove antecedent incompatibility $\vee$ exhibit the survivor.

Source familiarity, disciplinary preference, and requests for a different theorem class carry no evidential force without one of these witnesses.

Maximal positive conclusion

The strongest parent-scope conclusion is exact boundary normal-form homology plus no inheritance. For the certified black-hole target, residual closure fixes the endpoint token and classifies every unresolved burden; neither the metric interior nor its endpoint can authorize ANF, carry old content through the boundary, or establish the internal population of a successor domain.

Claims and nonclaims

The theorem package establishes a one-way dependency order from kernel and AMetric invariant to physical interiors; an origin-side initial-condition standing theorem; a licensed and bounded Einsteinian metric interior; an endpoint-side boundary-exposure theorem; equality of the complete AMetric boundary role profiles at token-distinct origin and endpoint positions; no crossing or inheritance; a target-class exact black-hole endpoint theorem; and a bivalent black-hole burden corollary. It does not claim that metric content authorizes the invariant, and it does not infer endpoint status from an astronomical label alone. It claims exactness for the certified post-subtraction target: the nonempty unresolved residue left after the declared exterior quotient and all genuine certificates have done their work.

1 Dependency-order lock and theorem object

1.1 One-way dependency order

The primitive AASC dependency order is

$$\begin{aligned} \mathcal{K} = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}) &\implies \text{ANF} \implies B_A^0 \\ &\implies \text{admitted physical interior} \implies \Pi_g \\ &\implies \text{endpoint-side exposure.} \end{aligned}$$

The first implication fixes the governance work required for non-degenerate identity-preserving construction. The second fixes the AMetric boundary normal form. Every later object is downstream: initial data, clocks, records, source ledgers, metric comparison, Einstein dynamics, horizons, black-hole burdens, residual coherence, and endpoint descriptions.

The reverse arrows are excluded:

$$\Pi_g \not\Rightarrow \mathcal{K}, \quad H_{\text{BH}} \not\Rightarrow \text{ANF}, \quad B_A^* \not\Rightarrow \text{ANF}.$$

A downstream object may instantiate the target class or satisfy the normal form. It may not authorize the conditions that made its own physical use well typed.

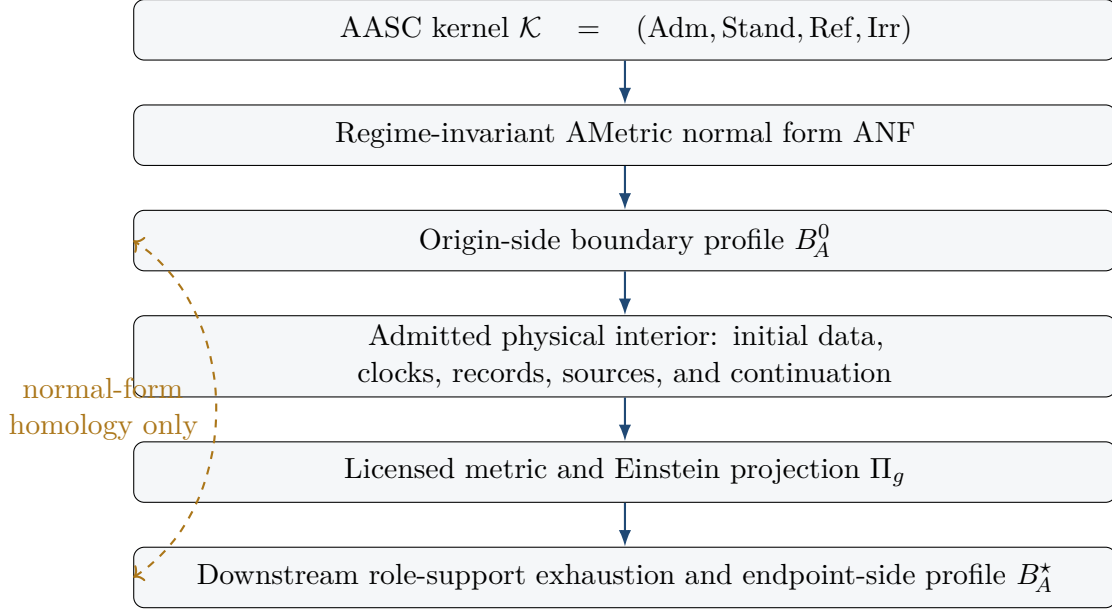


Figure 2: Dependency order and AMetric normal-form homology. The dashed relation compares boundary role-profiles; it is not a physical transition or an authorization arrow.

1.2 Correctness burden and dependency posture

The theorem begins after the AASC kernel and AMetric invariant have been fixed. Its correctness is therefore assessed by target preservation and dependency use, not by whether a critic prefers a different physical mechanism. A responsive objection must do at least one of the following:

- (C1) deny that the declared origin-endpoint package instantiates the fixed kernel-faithful target class;
- (C2) break a load-bearing kernel, AMetric, metricity, Einstein-projection, trajectory, residual-coherence, endpoint, no-crossing, or null-regime theorem;
- (C3) identify an invalid specialization or an overextended use of an imported result;
- (C4) establish that the capstone antecedents cannot jointly instantiate; or
- (C5) exhibit a standing-relevant same-scope boundary role, endpoint relation, transport route, selector, repair, or successor authority outside the stated normal forms.

The capstone survivor burden is precise: preserve the theorem object and produce a role-profile coordinate or lawful relation not fixed by the origin proof, endpoint exhaustion, independent gating, and no-crossing. A different theory, additional operator, common Hilbert space, new ledger, or new domain is a declared scope change unless it preserves the original comparison class.

Kernel-formalization import lock. This manuscript treats the kernel and AMetric invariant as upstream proof infrastructure, not as hypotheses generated by black-hole endpoint behavior. The

kernel layer is tied to the AASC non-degenerate construction formalization: admissibility, standing, reference, and irreversibility are used here as fixed-domain roles whose uniqueness and non-derivability-from-below status are imported from the kernel level. The present paper does not re-formalize those Lean results; it uses them to compare the origin-side and endpoint-side boundary role profiles without allowing metric, horizon, singularity, or endpoint structures to authorize their own governance.

1.3 No downstream authorization

Theorem 1.1 (No Downstream Authorization of the AMetric Invariant). *No metric, causal, temporal, source-ledger, horizon, singularity, black-hole, cosmological, or endpoint structure can authorize the AASC kernel or the AMetric normal form. Such structures may instantiate, project, or expose consequences of the invariant only after their own standing is fixed.*

Proof. Every listed structure presupposes target fixation, eligibility for use, redescription stability, and retained continuation. Metric objects additionally presuppose licensed comparison and transport; causal and temporal objects presuppose ordered dependence; source ledgers presuppose quantity identity and closure; horizons presuppose a metric/causal projection; endpoints presuppose a target whose role support can be tested. Using any of them to authorize the kernel or ANF would make a downstream role establish the conditions of its own admissibility. That is boundary backflow. The only lawful relation is application of the already fixed invariant to a downstream target. $\square$

Corollary 1.2 (Exposure, not production). *A downstream endpoint does not produce, form, generate, or select an AMetric boundary. It exposes a boundary-status already classified by the AMetric invariant when the relevant old-scope roles no longer have standing-bearing support.*

Proof. Production and formation require temporal order, a process carrier, and an admitted operator. Those roles are excluded at an AMetric boundary. Endpoint classification is therefore theorem-level normal-form satisfaction rather than physical boundary manufacture. $\square$

1.4 General and black-hole target packages

Definition 1.3 (General homology target). The general origin-endpoint theorem object is

$$\mathfrak{H}_A^{\text{gen}} = (\mathcal{K}, \text{ANF}, \mathcal{P}_{\text{ANF}}, B_A^0, P^*, \text{RouteClose}, \text{NullClose}, \text{NoCross}).$$

It records the dependency-prior kernel and normal form, the fixed role-profile coordinates, the origin token, an abstract downstream endpoint profile, route closure, null closure, and no-crossing. It contains no black-hole-specific structure and asserts no common geometry, physical transition, or inherited content.

Definition 1.4 (Certified black-hole homology target). The black-hole specialization is

$$\mathfrak{H}_A^{\text{BH}} = (\mathfrak{H}_A^{\text{gen}}, D_0, \Pi_g, H_{\text{BH}}, \mathcal{B}_H, \mathcal{W}_{\text{ext}}, R_H, \text{Succ}_{D_0}^{\text{res}}).$$

It adds the admitted parent domain, licensed metric/source projection, derivative horizon role, declared reconstruction burden, certified exterior-witness subtraction, unresolved residue, and residual-successor relation. Satisfaction of the endpoint-exhaustion certificate instantiates the abstract profile P^* as B_A^* . The specialization $\mathfrak{H}_A^{\text{BH}} \Rightarrow \mathfrak{H}_A^{\text{gen}}$ is a proof relation, not a physical arrow.

Definition 1.5 (Homology comparison class). A proposed variation remains same-scope for the general theorem only when it preserves:

- (H1) the kernel roles, bivalent standing discipline, irreversibility, and no same-act repair;

- (H2) the AMetric predicate, boundary signature, evaluated fragment, lawful quotient, retained continuation, and anchor–tensor–skin carrier;
- (H3) the role-profile coordinates, canonical vector $\mathbf{a}_{\text{ANF}}$, and token-distinction discipline;
- (H4) the origin and endpoint theorem positions; and
- (H5) route closure, null closure, independent target gating, and no-crossing.

A black-hole same-scope challenge must additionally preserve:

- (BH1) the parent domain, licensed metric/source envelope, and derivative horizon role;
- (BH2) the exterior-witness subtraction rule, nonempty unresolved residue, and old-scope standing-transport status; and
- (BH3) residual-support exhaustion and the absence of prior independent re-governance.

Changing a required item changes the theorem object, resolves or removes the target, denies certification, or enriches scope. A faithful challenge preserves the applicable package while carrying a standing-relevant difference.

1.5 Boundary profiles and role-profile identity

Let B_A^0 denote the origin-side AMetric boundary profile associated with admission of a standing-bearing physical interior. Let $B_A^\star$ denote an endpoint-side profile obtained when a downstream governed regime satisfies the endpoint normal form.

Definition 1.6 (Boundary role-profile map). For a boundary profile B , define

$$\mathcal{P}_{\text{ANF}}(B) = (\mathbf{M}_B, \mathbf{T}_B, \mathbf{C}_B, \mathbf{O}_B, \mathbf{S}_B, \\ \mathbf{X}_B, \mathbf{V}_B, \mathbf{R}_B, \mathbf{G}_B, \mathbf{A}_B).$$

The coordinates record, respectively: metric/comparison/ranking authority; temporal/order/process authority; causal-production authority; operator-domain authority; selector/probability/parameter authority; cross-domain standing, reference, law, and ledger transport authority; standing-valuation status; same-act repair or reversal authority; target-gating status; and self-authorization status. These coordinates are not additional structure introduced to force a homology conclusion. They are the operational expansion of the AMetric normal-form clauses into the boundary roles those clauses govern. With these coordinates fixed, set

$$\mathbf{a}_{\text{ANF}} = (0, 0, 0, 0, 0, 0, \text{bi}, 0, \text{ind}, 0),$$

where 0 means absence of the corresponding boundary authority, bi means bivalent standing on the evaluated fragment with undefinedness treated as non-membership, and ind means that every positive target requires independent gating. The normal-form predicate is represented by

$$\text{ANF}(B) \iff \mathcal{P}_{\text{ANF}}(B) = \mathbf{a}_{\text{ANF}}.$$

This equivalence is notation-fixing: $\text{ANF}(B)$ is the predicate-level statement, while $\mathcal{P}_{\text{ANF}}(B) = \mathbf{a}_{\text{ANF}}$ is its coordinate representation. The substantive proof burden is the independent derivation of this vector at each boundary token.

Definition 1.7 (AMetric normal-form equivalence). For profiles in the fixed comparison class, write

$$B_1 \cong_{\text{ANF}} B_2$$

iff

$$\mathcal{P}_{\text{ANF}}(B_1) = \mathcal{P}_{\text{ANF}}(B_2) = \mathbf{a}_{\text{ANF}}.$$

The relation is equality of admissibility-bearing boundary roles, not equality of adjacent contents, locations, histories, or causal ancestry.

Normal-form homology terminology lock

“Homology” means equality of independently derived normal-form role profiles. It does not mean algebraic-topological or chain homology, causal ancestry, temporal recurrence, common geometry, shared state space, or inherited content. The comparison relation is external to the boundary tokens and reports only $\mathcal{P}_{\text{ANF}}(B_1) = \mathcal{P}_{\text{ANF}}(B_2) = \mathbf{a}_{\text{ANF}}$.

Definition 1.8 (Token distinction). $B_1 \neq B_2$ means the boundary profiles occupy distinct theorem positions or belong to distinct governed domain interfaces. Role-profile equality never implies token identity, temporal return, causal relation, common geometry, or inherited content.

Lemma 1.9 (Profile Equality Is Not Predicate Repetition). *AMetric normal-form homology is not established by assigning the same name to two boundary positions. It is established only when every coordinate of $\mathcal{P}_{\text{ANF}}$ is independently fixed at each token and the resulting vectors are equal to $\mathbf{a}_{\text{ANF}}$.*

Proof. A profile may be nonmetric while retaining probability, ordering, operatorhood, transport, a selector, partial-standing language, repair, or self-authorization; such a profile does not equal $\mathbf{a}_{\text{ANF}}$. The equivalence $\text{ANF}(B) \iff \mathcal{P}_{\text{ANF}}(B) = \mathbf{a}_{\text{ANF}}$ only fixes notation; it does not supply either boundary proof. The origin proof fixes its coordinates from pre-admission closure. The endpoint proof fixes its coordinates from downstream role-support exhaustion and route exclusion. Equality therefore compares independently established role statuses rather than repeating a label. $\square$

The main target is

$$B_A^* \neq B_A^0, \quad \mathcal{P}_{\text{ANF}}(B_A^*) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}, \quad B_A^* \cong_{\text{ANF}} B_A^0.$$

The certified black-hole package supplies a target-class certification of B_A^* ; it does not define ANF, authorize B_A^0 , or convert endpoint classification into a physical transition.

1.6 Exact homology-theorem dependency map

The local result supplied by each dependency is fixed before the capstone comparison is made.

Table 1: Exact dependency map for the normal-form homology theorem.

Homology-specific result	Imported closure object
Origin-side AMetric profile	Admissibility-structure and AMetric boundary theorems: pre-boundary AMetricity, bivalence, standing identity, comparison-class lock, independent target gating, and no self-authorization.
No primitive initialization	Initial-condition standing, AMetric no-selector closure, and anchor–tensor–skin role exhaustion.

Homology-specific result	Imported closure object
Licensed physical interior	Realized-interior theorem, metricity licensing, quotient stability, clocks, records, source continuation, and no boundary backflow.
Einstein metric/source envelope	AASC gravity role and the minimal first-response same-scope Einstein projection; no universal-gravity claim is imported.
Endpoint role-profile derivation	No horizon-as-AMetric-reset, residual coherence, old-scope role-support exhaustion, coordinate population, false-endpoint exclusion, null-route closure, and no self-relicensing.
Token distinction	Standing monotonicity, irreversibility, no same-act repair, and no-crossing.
Normal-form homology	Independent origin and endpoint coordinate derivations followed by vector equality.
No cyclicity and no inheritance	AMetric non-transmissivity, independent target gating, and absolute no-crossing.
Black-hole burden bivalence	Kernel bivalence and certified exterior-witness discipline, applied after the homology theorem's invariant is fixed.

1.7 Non-circularity

The theorem assumes no black-hole endpoint, successor domain, cyclic map, information-transfer channel, metric-singularity authority, or populated post-boundary domain. It imports the kernel and AMetric invariant as dependency-prior results. The local work is to derive the origin profile, derive the endpoint profile under role-support exhaustion, prove token distinction, and compare the independently populated vectors. No clock, metric, initial datum, black-hole residue, singularity, or endpoint theorem is used to establish the kernel or the invariant that licenses its own use.

1.8 Counterexample conditions

The theorem is AASC-internal and dependency-locked. A same-scope counterexample must deny target-class certification, break an imported theorem, identify a misuse of an import, or exhibit a faithful survivor that preserves the comparison class while escaping the stated role-profile and route-exhaustion results. Adding a new metric, selector, clock, probability measure, boundary channel, common Hilbert space, law package, or external meta-domain changes the theorem object unless the original comparison class is preserved.

1.9 Binding theorem ledger

Table 2: Binding dependencies in primitive-to-application order.

Imported result	Exact imported work	Role in the theorem package
AASC kernel	Necessity of admissibility, standing, reference, irreversibility, target identity, and no same-act repair (Maley, 2026g).	Fixes the dependency-prior governance package.
Admissibility structure	AMetric proposal-side collapse; bivalence; standing as reuse-stable admissibility; unique admissible interior; conservation and no same-scope creation or repair of standing (Maley, 2026n).	Fixes the gate-bearing status architecture used by both boundary profiles.
AMetric boundary theorem	Comparison-class lock; no-selector/no-parameter closure; behavior exhaustion; route collapse; independent target gating (Maley, 2026i).	Supplies the invariant normal form and export discipline at both boundary tokens.
Unique fixed-domain interior	No competing same-domain gates or plural admissible interiors (Maley, 2026p).	Blocks rival boundary governance while preserving possible independently gated domains.
Initial-condition and standing-monotonicity results	No privileged primitive start and initial conditions as anchors (Maley, 2026b,e); retained-consequence orientation (Maley, 2026j).	Supplies the origin-side physical-data application and no-reversal discipline.
Realized interior and role decomposition	Downstream physical role ladder and anchor-tensor-skin exhaustion (Maley, 2026a,l).	Types initial data, metric/source content, witness maps, and skin.
Metricity, gravity, and Einstein projection	Metricity as licensed description (Maley, 2026f); GR-type constraint role and minimal first-response Einstein metric/source projection (Maley, 2026c,d).	Fixes the downstream governed metric interior whose application domain may later exhaust.
Residual coherence and black-hole endpoint	Derivative-horizon/no-reset discipline, horizon burden, exterior witness subtraction, residual coherence, governance-depth, New AMetric Boundary exactness, and no-crossing (Maley, 2026h,k).	Supplies the certified black-hole endpoint instantiation and application corollary.
Null-regime closure	Exclusion of admitted total vacuity (Maley, 2026m).	Closes the null endpoint route.

2 The AMetric invariant and origin-side boundary normal form

2.1 Origin boundary versus initial condition

The AMetric invariant is dependency-prior to every physical initial datum and every metric-interior construction. It is not a first state, first instant, zero-size geometry, quantum vacuum, law package, probability distribution, or selected member of a family. It is the boundary-status at which target identity and eligibility can first be fixed without presupposing an admitted physical comparison structure.

Definition 2.1 (Origin-side AMetric boundary). B_A^0 denotes the AMetric boundary profile relative to the admission of a standing-bearing physical interior. The superscript zero marks origin-side theorem position, not temporal coordinate zero.

Define $\text{ANF}(B)$ by the conjunction of:

- (N1) no boundary metric, order, ranking, probability, typicality rule, threshold, index, scale, proximity relation, or canonical representative;
- (N2) no event, execution, transformation, production, or process at the boundary;
- (N3) no admissible operator with the boundary as an inherited transformation domain;
- (N4) no transport of prior standing, reference, law, metric, clock, source ledger, or continuation authority;
- (N5) bivalent standing only on the evaluated fragment, with undefinedness as non-membership rather than a third value;
- (N6) no same-act repair;
- (N7) independent target gating for every positive target domain; and
- (N8) no metric, temporal, causal, modal, probabilistic, physical, or formal structure authorizing its own admission.

Theorem 2.2 (Origin-Side Boundary Theorem).

$$\text{ANF}(B_A^0), \quad \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}.$$

Proof. Before fixation, no representative has standing as this target rather than another. Any pre-fixational asymmetry requires an individuator, selector, order, metric, probability, or canonical representative, all excluded by the AMetric theorem. Boundary action and operatorhood presuppose the licensing they would have to supply. Cross-domain standing transfer would use a source gate to govern a target without independent closure. Bivalence, no-repair, independent gating, and no self-authorization follow from the kernel and the AMetric theorem. These clause-level results populate every coordinate of $\mathcal{P}_{\text{ANF}}(B_A^0)$ with the corresponding entry of $\mathbf{a}_{\text{ANF}}$ (Maley, 2026i,n). $\square$

Theorem 2.3 (No Primitive Initialization). *No physical state, field value, low-entropy macrocondition, probability distribution, vacuum state, metric geometry, causal relation, law package, quantum state, or cosmological boundary value can be the primitive initial condition of a standing-bearing physical interior. Such objects may be admitted downstream as anchors or tensors; they cannot occupy B_A^0 without violating ANF.*

Proof. Each object contains post-admission structure. A state needs a target state space and identity relation; a metric needs comparison and transport; probability needs events and measure; a field needs a carrier; causality needs order and witnessed dependence; a law needs a domain and operators; entropy needs macrostate partition and comparison. Used downstream, these are legitimate. Used as boundary authorizers, they import the structure excluded by ANF. $\square$

Corollary 2.4 (Initial conditions are first licensed interior roles). *What ordinary physics calls an initial condition is, at the earliest admissible position in the role order, an interior-side anchor or tensor downstream of boundary admission. It is never identical with the AMetric boundary.*

2.2 Comparison-class lock and behavior exhaustion

A boundary proposal remains same-type only when it preserves the boundary signature, evaluated fragment, standing partition, lawful quotient, finite-history continuation structure, and anchor–tensor–skin role carrier. Altering one of these changes the theorem object.

Within that class, constructional behavior is exhausted by evaluated-act classification and coherent retained-trajectory structure. Admission, rejection, and fixation occur at the first locus; reuse, record retention, and continuation occur at the second. A third claimed locus either changes one of these, changes scope, or is bookkeeping.

Lemma 2.5 (No Hidden Boundary Locus). *Any distinction relevant to initial-data standing or origin-endpoint normal-form homology alters evaluated classification or retained continuation. A distinction changing neither cannot distinguish B_A^0 from $B_A^\star$ as AMetric boundary types.*

2.3 Per-domain uniqueness

The AMetric and *Unus* results establish uniqueness of the admissible interior for a fixed standing-governed domain, including elimination of competing gate-induced interiors (Maley, 2026i,p). This is per-domain uniqueness. It does not imply that only one domain can exist or that a new independently gated domain inherits the first.

3 Initial conditions as downstream anchor, tensor, and certificate roles

3.1 Origin-side application: primitive starts and admitted data

The origin-side boundary theorem fixes what initial conditions cannot be. No physical datum sits on the pre-admissible side of B_A^0 . Initial data enter only after the AMetric invariant has licensed a standing-bearing interior and the relevant comparison package has been declared.

3.2 Primitive starts and admitted data

Definition 3.1 (Primitive-start claim). A primitive-start claim is a purported global configuration whose privilege as beginning, origin, low-entropy condition, vacuum state, field value, or cosmological selection is used to explain or authorize the physical regime in which that privilege is stated.

Definition 3.2 (Admitted initial datum). An admitted initial datum is a datum, class of data, or boundary condition inside an already licensed model. Its target, domain, constraint status, redescription behavior, source and clock roles where applicable, continuation rule, and failure conditions are declared independently of its numerical or representative value.

The first is excluded by AMetric boundary closure. The second is ordinary physical practice. AASC rejects the boundary-authorizing role while preserving the admitted-data role.

Proposition 3.3 (Primitive/admitted distinction). *There is no conflict between rejecting a privileged global initial condition as a primitive physical input and admitting ordinary initial data as anchors or tensors of a declared model.*

Proof. The primitive-start claim attempts to determine the target, metric, temporal, probabilistic, or causal structure required to formulate it. It performs boundary-authority work. An admitted datum acts after the comparison class is fixed and may alter solutions without authorizing the boundary. The roles are different: selector versus declared anchor or tensor. $\square$

3.3 The initial-data certificate

Define an initial-data certificate for domain D and use U by

$$\mathcal{J}_D^U = (D, U, \Sigma, \mathcal{C}_D, \mathcal{Q}_D, \text{Ord}_D, \text{Source}_D, \text{Rec}_D, \text{Cont}_D, \text{Fail}_D),$$

where Σ is the declared initial or Cauchy locus, $\mathcal{C}_D$ the constraint package, $\mathcal{Q}_D$ the physical quotient, Ord_D the clock or ordering role, Source_D the source ledger where applicable, Rec_D the record/readout structure, Cont_D the continuation relation, and Fail_D the failure boundary.

Definition 3.4 (Certified initial datum). A datum is certified when its target is fixed under lawful redescription; its anchor or tensor role is explicit; it satisfies the active constraints and descends to the physical quotient; its initiality is well typed by the declared order; its source, record, and continuation effects are included; its failure conditions are explicit; and no undeclared selector, probability, typicality, ranking, convenience, or desired outcome supplies its privilege.

Theorem 3.5 (Initial-Condition Standing Theorem). *A purported initial condition has physical standing only as a certified datum inside an initial-data package. It cannot primitive-authorize the metric, temporal, causal, probabilistic, source, record, or continuation structures required for its own identification and use.*

Proof. Physical use requires target fixation, lawful variation, quotient identity, and continuation. Metric or Cauchy data also require geometric and constraint roles. Low-entropy data require state space, coarse-graining, measure, and records. Scalar-field or vacuum data require field, source, and calibration roles. If those roles are declared, the datum is an anchor or tensor. If they are not and the datum still selects a start, it acts as a boundary selector, excluded by AMetric no-selector closure. Adding missing roles after a desired history is known is repair; adding them by changing the theorem object is scope enrichment. No same-scope primitive route remains. $\square$

Corollary 3.6 (Ordinary initial-value practice is preserved). *A model may legitimately take initial data as supplied input and derive conditional consequences. Derivation from initial data does not by itself predict, explain, or primitive-authorize the data.*

Proposition 3.7 (Origin-Side Profile Fixation). *Because every physical initial datum requires target, quotient, order, metric/source, record, and continuation roles before it can function as a datum, no such datum contributes a boundary-authorizing coordinate to $\mathcal{P}_{\text{ANF}}(B_A^0)$. Initial data therefore enter only on the admitted side of B_A^0 .*

Proof. Each datum is typed by roles that are absent as primitive boundary authority in $\mathbf{a}_{\text{ANF}}$. Once those roles are declared, the datum is an anchor, tensor, or certificate. If the datum is used to supply those roles, it acts as an excluded selector or self-authorizer. No intermediate role exists in the fixed comparison class. $\square$

This distinction aligns with the ordinary GR initial-value problem: well-posed development from data satisfying the constraints is not a theorem selecting those data (Choquet-Bruhat and Geroch, 1969; Wald, 1984).

Initial-data element	AASC role	Standing-bearing work
Choice of hypersurface, boundary class, or admissible data sector	Anchor	Fixes definedness, boundary typing, or the comparison domain for the initial-value problem.
Constraint-satisfying metric, extrinsic-curvature, matter, or gauge-invariant data	Tensor	Carries quotient-level physical content inside the licensed metric/constraint package.
Coordinate chart, component list, foliation label, or representative notation	Skin unless independently role-fixed	Presents admitted data but carries no physical privilege solely by representation.
Low-entropy macrocondition or Past-Hypothesis-style boundary assumption	Anchor	Fixes a cosmological or statistical boundary condition after state space, coarse-graining, measure, and record roles are declared.
Scalar-field or vacuum datum in a declared model	Tensor or anchor, according to use	Acts as physical state content when varied inside fixed anchors; acts as an anchor when it fixes the active sector or boundary package.
Typicality, simplicity, convenience, or fit used to choose one start	Unsupported selector unless declared as richer scope	Cannot acquire primitive boundary authority from ranking or performance alone.

Table 3: Initial data do not share one undifferentiated role. Their status depends on the standing-bearing work performed in the declared model.

3.4 Past Hypothesis and inflationary data

A Past-Hypothesis-style low-entropy condition may be a powerful cosmological anchor, but it presupposes a state space, macrostate partition, measure, coarse-graining, transport, and record structure (Albert, 2000; Carroll, 2010; Maley, 2026j; Price, 1996). It may orient effective thermodynamic and record arrows inside a model; it cannot supply the standing conditions that make it a boundary condition.

Inflationary initial states make the same point. A scale factor, Hubble rate, scalar field, potential, vacuum phase, e-fold count, and reheating channel are downstream roles. They may realize an early transport/closure certificate, but none is pre-metric boundary authority (Guth, 1981; Maley, 2026b).

Theorem 3.8 (Initial-Condition Firewall). *No boundary datum can authorize the standing of the clock, metric, source, record, quotient, comparison, or continuation structure required to state that datum as a physical initial condition. Any attempted authorization is selector import, parameter injection, representative privilege, deferred admissibility, repair, scope enrichment, or bookkeeping.*

Proof. A physical initial condition must be distinguishable and reusable. By AMetric behavior exhaustion, every standing-relevant effect enters evaluated classification or retained continuation. At the first locus, primitive privilege requires selector, parameter, second classifier, or representative. At the second, it requires deferred admission, repair, or a tensor reified as boundary operator. Changing the comparison class is scope enrichment; changing neither locus is bookkeeping. $\square$

Initial-data audit

For every proposed initial condition ask: What target and use are fixed? What makes it the same datum under redescription? Is it anchor, tensor, or skin? What constraints and quotient apply? What clock or order makes it initial? What source, record, continuation, and failure ledgers are active? Is any probability, typicality, simplicity, or desired outcome doing undeclared selection work?

4 Licensed physical interior, Einstein projection, and standing monotonicity

The physical interior does not authorize the kernel or AMetric boundary. Its role is downstream realization: once standing-bearing domainhood is admitted, metric, source, clock, record, and causal structures may be licensed as interior content.

Theorem 4.1 (Licensed Physical Interior Theorem). *Every metric, source, clock, record, causal, and dynamical structure used in this theorem package is downstream of the kernel and AMetric invariant. Its physical standing is conditional on the already fixed domain, quotient, continuation, and failure structure; it carries no boundary-authorizing backflow.*

Proof. Each structure changes either quotient-level physical content or the definedness of lawful continuation inside an admitted domain. It therefore occupies an anchor or tensor role after admission. Treating it as prior authority would reverse the dependency order established by the kernel and the AMetric theorem. $\square$

4.1 Metricity is licensed, not primitive

The imported metricity theorem defines an L-metric boundary as a local projection interface at which metric descriptors become physically interpretable in a dependency order. Metric descriptors are licensed only as standing-preserving interior articulation; they are not boundary authority, primitive selectors, boundary operators, transmission channels, or repair devices (Maley, 2026f).

Theorem 4.2 (Metric Licensing Theorem, imported). *Any physically standing-bearing metric descriptor is licensed only downstream as interior articulation of already standing-positive structure. No metric fact retro-authorizes the boundary standing of the regime in which it is used.*

A Lorentzian model may introduce manifold and metric jointly inside an admitted model. What is excluded is using that jointly introduced metric as the pre-admissible gate.

4.2 The AASC gravity role and Einstein projection

The imported AASC gravity theorem identifies gravity at the role level as the constraint architecture required for an admissible physical interior: standing surface, observable quotient, admissible redescription, first-class closure, and multiplier-type Hamiltonian presentation (Maley, 2026d). Curvature is a faithful projection of that role once metric description is licensed.

Definition 4.3 (Minimal first-response same-scope metric projection). A metric/source projection belongs to the minimal first-response same-scope class when it is four-dimensional and Lorentzian; local, symmetric, and rank two; divergence-compatible with the declared closed source ledger; limited to the first curvature-response basis; and introduces no independent higher-response tensor, extra propagating field carrier, nonlocal operator, or enlarged comparison scope.

Let $\Pi_g(\mathcal{A}_G) = (M^4, g_{\mu\nu})$ be a faithful projection in this class. By definition of the first-response basis, the geometric response is exhausted by

$$R_{\mu\nu}, \quad Rg_{\mu\nu}, \quad g_{\mu\nu},$$

so

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

Closed source continuation gives $\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$, hence $\nabla^\mu E_{\mu\nu} = 0$. Metric compatibility and the contracted curvature identity fix $b = -a/2$, yielding

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Theorem 4.4 (Licensed Einstein Projection, imported). *Within a faithful four-dimensional Lorentzian metric projection of the AASC gravity role, and under the minimal first-response same-scope predicate, Einstein dynamics is the unique continuation-compatible metric field-equation form. It has standing as a projected theorem object, not primitive ontology.*

The result is consistent with metric-internal uniqueness classifications such as Lovelock’s theorem while carrying a different upstream burden: licensing metricity, typing the source ledger, and controlling same-scope additions (Lovelock, 1971; Maley, 2026c). The capstone homology theorem requires a licensed downstream metric/source envelope whose role support can exhaust; the Einstein form supplies the declared physical envelope for the black-hole specialization, not the source of AMetric homology.

Remark 4.5 (Scope of the projection theorem). Higher-curvature, quantum, nonlocal, effective, or extra-field extensions are not denied. They lie outside the minimal first-response same-scope class unless their added tensors, carriers, or operators are shown to reduce to its declared basis without enlarging the theorem object. The homology theorem does not require Einstein projection as a universal theory of gravity; it requires a licensed downstream metric/source envelope whose role-support can be tested for exhaustion.

4.3 Einsteinian initial data

Define a governed metric-interior package

$$\mathfrak{M}_D = (M, g, \nabla, \mathcal{Q}_D, T_{\mu\nu}^{\text{closed}}, \text{Clock}_D, \text{Rec}_D, \text{Cont}_D, \text{Fail}_D).$$

Einsteinian Cauchy data have standing only relative to this package and the active gravitational constraints. A maximal development can be unique in the appropriate mathematical sense while the selection of the initial data remains a declared anchor, empirical input, or richer-scope result (Choquet-Bruhat and Geroch, 1969; Wald, 1984).

The source tensor is the metric-facing closed ledger of admitted deformation response, field carriers, particle persistence, transport, interactions, restoration burden, and homogeneous vacuum content. Its covariance and divergence compatibility do not make it a cross-AMetric substance.

4.4 Standing monotonicity and the meaning of initiality

Let E_D be standing-bearing stages and let $\equiv_S$ identify stages with the same retained standing content. The quotient $\mathcal{Q}_S(D) = E_D / \equiv_S$ carries a partial order of retained consequence. Its strict part gives standing monotonicity: an irreversible standing-bearing fixation leaves a downstream difference that cannot be erased as the same lineage (Maley, 2026j,o).

Definition 4.6 (Standing-relative initiality). A datum or stage is initial for a declared problem when it occupies the anchor position from which the relevant continuation, prediction, reconstruction, or record lineage is evaluated. Initiality is relative to the use and comparison class.

Proposition 4.7 (Initiality is not metaphysical firstness). *Standing-relative initiality does not imply the first event in an absolute time, the first state of reality, or a pre-admissible cause. A Cauchy surface can be initial for one development problem while embedded in a larger spacetime; a low-entropy state can be initial for one statistical explanation while depending on a broader cosmological model.*

Proposition 4.8 (Standing-Monotonicity Role in the Homology Theorem). *Standing monotonicity is not an additional premise in the derivation of $\text{ANF}(B_A^*)$. Its exact role is interpretive and eliminative: it fixes initiality and endpoint as positions in a retained standing-bearing continuation order, and it excludes temporal reversal, cyclic return, or restoration of an earlier standing state as readings of normal-form recurrence.*

Lemma 4.9 (No Temporal Return from Normal-Form Homology). *No extension of a parent standing-bearing lineage can reach B_A^* and count as a return to B_A^0 while preserving the same lineage.*

Proof. A same-lineage return would require erasing the admitted metric, temporal, source, record, and operator roles as though they had never been fixed, violating standing monotonicity and no-repair; or carrying them through the boundary, violating non-transmissivity; or declaring a fresh domain, which is scope reset rather than return. $\square$

Thermodynamic, record, causal, quantum-measurement, psychological, and cosmological arrows remain effective projections or refinements over the standing order. The recurrence theorem does not reverse any of them; it terminates the old scope’s authority to apply them across the endpoint.

5 Endpoint-side boundary exposure

Endpoint-side exposure is first stated at the role-profile level: when the support of a licensed regime exhausts, which boundary profile remains admissible? The black-hole residue supplies the certified target-class certification because it fixes a licensed metric regime, a derivative horizon role, a reconstruction burden, exterior-witness subtraction, and an exact test for residual metric-support exhaustion without treating GR as primitive ontology.

Lemma 5.1 (Endpoint Coordinate Population). *Let P^* be an endpoint profile in $\mathfrak{H}_A^{\text{gen}}$. If the old-scope role supports and rival routes listed in table 4 are exhausted, every coordinate of $\mathcal{P}_{\text{ANF}}(P^*)$ is fixed independently. No coordinate is assigned by applying the name “AMetric” to the endpoint.*

Table 4: Independent population of the endpoint AMetric role-profile coordinates.

Coordinate	Independent endpoint condition	Closed same-scope rival
M: metric/-comparison	Residual metric-comparison successor support is empty for the endpoint target.	Ordinary GR interior, metric depth, affine continuation, or classical singularity as final authority.
T: tempo-ral/process	No old-scope retained-continuation order or process domain remains for the target.	Destruction event, bounce, transition, temporal return, or endpoint process.

Coordinate	Independent endpoint condition	Closed same-scope rival
C: causal production	No common old-scope order and standing-bearing witness relation spans parent and successor.	Causal creation, production of a successor, or cross-boundary influence.
O: operator domain	The old operator role has no endpoint domain and no replacement operator has been independently governed.	Same-scope tunneling, crossing, execution, or transformation at the boundary.
S: selector/-parameter	AMetric no-selector/no-parameter closure excludes ranking, probability, branch index, threshold, and representative privilege.	Preferred successor, probabilistic selection, typicality, or canonical branch.
X: standing transport	Old-scope standing transport is undefined; no carrier, storage relation, or inherited ledger is available.	Inherited information, energy, law, identity, record, or source package.
V: valuation	Bivalence governs the evaluated fragment; undefinedness is non-membership rather than a third standing value.	Weak, latent, partial, conserved, or probabilistically positive standing.
R: repair	Irreversibility and no same-act repair remain binding after standing loss.	Remnant, archive, reconstruction, or same-lineage restoration as repair.
G: target gating	No parent gate crosses; every positive post-boundary target requires independent governance.	Automatic successor admission or inherited law package.
A: self-authorization	No exhausted metric, equation, endpoint, law, or ledger may re-license its own application.	Endpoint backflow or self-relicensing.
Endpoint null closure	Admitted total vacuity is unavailable; mute nullity is AMetric pre-admissibility and performs no positive classification.	“Nothingness” as a same-type endpoint rival.

Proof. Each condition concerns an independently typed role whose standing support can be tested before the endpoint is named. A surviving rival would require the authority shown in the third column. Exhaustion of that authority fixes the associated coordinate. The metric coordinate follows from residual metric-support exhaustion; the temporal, causal, and operator coordinates follow from loss of their old-scope domains; transport follows from undefined standing continuation; valuation and repair follow from bivalence and irreversibility; gating and self-authorization follow from independent target admission and no downstream authorization; and null closure follows from the inadmissibility of an admitted vacuous regime. The populated vector is therefore derived rather than stipulated. $\square$

Theorem 5.2 (Endpoint-Side Boundary Exposure Theorem). *Let $P^\star$ be the endpoint profile of a downstream regime governed by the AASC kernel. Suppose the old-scope metric, temporal, causal, source-ledger, record, and operator roles have no remaining standing-bearing support for the target; no independent re-governance has been declared; and selector import, repair, inherited transport, storage, dynamical transition, representative privilege, and admitted nullity are excluded. Then*

$$\text{ANF}(P^\star), \quad \mathcal{P}_{\text{ANF}}(P^\star) = \mathbf{a}_{\text{ANF}}.$$

The theorem applies the invariant; it does not derive the invariant from the exhausted regime.

Proof. By the Endpoint Coordinate Population Lemma, every coordinate of $\mathcal{P}_{\text{ANF}}(P^\star)$ is independently fixed to the corresponding entry of $\mathbf{a}_{\text{ANF}}$. The endpoint therefore satisfies the already fixed normal-form predicate. No downstream role contributes authority to the invariant used in the classification. $\square$

5.1 Certified black-hole target: the unresolved residue

The black-hole endpoint theorem is universal over its certified AASC target class. It does not classify every mathematical interior and does not depend on a literally permanent horizon. Its target is a parent domain D_0 carrying the licensed Einstein projection and a declared reference-standing reconstruction problem. Certification determines whether that target has a nonempty unresolved residue at maximum governance-depth; it does not make the endpoint optional once the target is fixed.

Let $\mathcal{B}_H$ be the family of parent reference-standing burdens generated when declared distinctions become dependent on a horizon-isolated carrier. Let $\mathcal{W}_{\text{ext}}(\mathcal{B}_H)$ be the portion genuinely resolved by standing-bearing exterior witness relations. Define

$$R_H = \mathcal{B}_H \setminus \mathcal{W}_{\text{ext}}(\mathcal{B}_H).$$

The notation is a role ledger, not a set of primitive information tokens.

Definition 5.3 (Standing-bearing exterior witness relation). An exterior relation counts as a witness only when it fixes a target reference class, supplies an exterior carrier or record family, preserves the quotient under lawful redescription, states success and failure conditions, remains stable under continuation, and does not use hidden interior storage as the source of identity.

For a burden $b \in \mathcal{B}_H$, a positive witness has the schematic form

$$w_b : b \mapsto \hat{b}_{\text{ext}},$$

where the codomain object is carried by the declared exterior record or algebra, the target reference class is fixed, lawful redescription preserves the identification, continuation preserves reuse, and success and failure conditions are explicit. A Page curve or radiation flux by itself is a ledger. An explicit quotient-preserving reconstruction map can be a witness. The burden is removed from R_H only in the second case.

Radiation flux, entropy curves, Penrose diagrams, boundary slogans, and global-state labels are valuable ledgers or presentations when their declared roles are fixed. They reduce R_H only through an explicit target-fixing witness map. The map, not the slogan or curve, performs reconstruction.

Proposition 5.4 (Residue exactness). *A burden re-fixed by a genuine exterior witness relation is removed from R_H . Exterior resolution does not repair the interior carrier or alter the endpoint normal form of any nonempty remainder.*

5.2 Derivative horizon and no AMetric reset

Theorem 5.5 (No Horizon-as-AMetric-Reset, imported). *A same-scope black-hole horizon H_{BH} is a derivative metric boundary whose identity depends on already licensed metric comparison, causal typing, exterior/interior role structure, and reconstruction burden. It cannot function as the same-scope AMetric licensing boundary for those roles. Treating it as a source of new standing is exactly declared new governance, a certificate, richer scope, hidden governance import, post-hoc repair, or bookkeeping (Maley, 2026h).*

Proof. The AMetric boundary is dependency-prior to metricity and comparison. A black-hole horizon is identified only inside an already licensed metric and causal projection. A structure that presupposes those roles cannot same-scope authorize them. Retyping the horizon as admissibility-generating therefore changes the theorem object or supplies an independently declared bridge; ordinary horizon crossing does not perform that work. $\square$

The horizon initiates the reconstruction burden; it is neither the endpoint token nor a replacement origin. Endpoint status is fixed only after exterior-witness subtraction and exhaustion of residual metric support for the unresolved residue.

5.3 Residual coherence and maximum governance-depth

For burdens retained in R_H , old-scope standing transport is undefined. Organized patterns may remain. Residual coherence classifies these patterns by role without restoring standing (Maley, 2026h).

Let

$$\text{Succ}_{D_0}^{\text{res}}(x, r)$$

denote post-break continuations retaining enough inherited organization for role r to remain descriptively usable without restoring old-scope standing.

Definition 5.6 (Maximum governance-depth). Maximum governance-depth for metric comparison is the status

$$\text{Succ}_{D_0}^{\text{res}}(R_H, \text{Metric}) = \emptyset.$$

It is not a radius, proper time, affine parameter, curvature threshold, or event. It is exhaustion of residual support for the metric role as applied to the unresolved residue.

Near a large horizon, residual metric and source organization may remain excellent. A freely falling description may be smooth and curvature modest. None of those facts supplies old-scope standing transport. Maximum governance-depth concerns the role-order exhaustion of residual authority, not local drama at the horizon.

Certified Black-Hole Endpoint Target

The black-hole specialization instantiates $\mathfrak{H}_A^{\text{BH}}$ exactly when the following coordinates are reported and satisfied:

1. the parent domain D_0 and its licensed metric/source projection Π_g ;
2. the declared horizon-generated burden $\mathcal{B}_H$;
3. the certified exterior-witness rule $\mathcal{W}_{\text{ext}}(\mathcal{B}_H)$;
4. a nonempty unresolved residue $R_H = \mathcal{B}_H \setminus \mathcal{W}_{\text{ext}}(\mathcal{B}_H)$;
5. undefined old-scope standing transport for every retained burden;
6. residual coherence distinguished from standing and from a hidden carrier;
7. empty residual metric successor support, $\text{Succ}_{D_0}^{\text{res}}(R_H, \text{Metric}) = \emptyset$;
8. absence of independent residue-level re-governance before endpoint classification;
9. closure of metric, temporal, causal, operator, carrier, selector, repair, storage, and inherited-law routes;

10. exclusion of admitted total vacuity, with mute nullity carrying no positive classificatory force; and
11. independent target gating and no-crossing for every positive post-boundary domain.

Failure of a required coordinate denies certification, resolves the residue before endpoint classification, or changes scope. It is not an alternative ending of the same certified unresolved residue and does not alter the capstone inference on a certified target.

5.4 Three levels of failure

Table 5: Equation, projection, and boundary levels.

Level	Typical description	AASC classification
Equation-level	Curvature divergence, geodesic incompleteness, inextendibility, high-curvature breakdown, or incomplete source response inside a metric theory.	Metric-internal diagnostic; may motivate declared higher-response or richer-scope structure.
Projection-level	Licensed metric projection loses residual comparison, source, measurement, or continuation support for the crossed target.	Exhaustion of the old metric role's application domain.
Boundary-level	No old-scope metric, time, causality, operator, carrier, source ledger, or transport authority remains; admitted nullity is unavailable.	New AMetric Boundary closure-status.

A classical singularity belongs to the first level. The recurrence theorem concerns the third, reached only after the projection-level antecedent is fixed.

5.5 False endpoint exhaustion

Any endpoint claim for the unresolved residue must be metric-bearing; process/temporal/causal-bearing; carrier/reference/source-ledger-bearing; null/vacuous; boundary/pre-admissible; or external third-domain comparison/bookkeeping. No seventh same-type profile remains inside the fixed comparison class.

Theorem 5.7 (False Endpoint Exhaustion). *Assume $R_H \neq \emptyset$, old-scope standing transport is undefined for burdens retained in R_H , no independent residue-level governance is declared before metric-role exhaustion, and residual metric successor support is empty. Then the endpoint is not admissibly classifiable as an ordinary same-scope GR interior, classical singularity, remnant, archive, information-destruction event, information-transfer endpoint, energy sink, dynamical bounce, wormhole throat, inherited-law transition, or admissible null regime.*

Proof. An ordinary GR interior and singularity require metric or affine diagnostic authority. A remnant or archive requires a standing-bearing carrier, encoding, identity, and retrieval map. Destruction and transfer require time, process, and operator. A bounce requires before/after order and dynamics.

A wormhole requires common geometry or larger path-integral scope. Inherited laws require cross-domain transport. An admitted null regime is contradictory because admission requires non-vacuous gate, identity, and continuation roles; mute nullity performs no endpoint classification and collapses to pre-admissible boundary status. These exhaust the metric, dynamical, carrier, transport, and null routes. $\square$

5.6 Null-route closure at endpoint

Theorem 5.8 (Endpoint Null-Route Closure, imported). *Total AASC vacuity cannot occupy the admitted endpoint side of a fixed-domain evaluability regime. An admitted endpoint requires target identity, gate status, standing-bearing classification, quotient stability, and continuation discipline; total vacuity supplies none. Completely mute nullity is AMetric pre-admissibility and has no positive endpoint force. If it is used to classify, select, exempt, or govern the endpoint, it imports non-null structure and is no longer mute (Maley, 2026m).*

Proof. Admitted nullity is contradictory because positive-side admission carries non-vacuous governance roles. Mute nullity supplies no target, comparison, selector, transport, or classification and therefore cannot rival a positive endpoint status. Any attempt to make it govern supplies precisely the non-null roles that vacuity denies. The null route is closed without instantiating successor content. $\square$

Theorem 5.9 (Black-Hole AMetric Endpoint Exactness). *Let a black-hole package instantiate $\mathfrak{H}_A^{\text{BH}}$: $R_H \neq \emptyset$, old-scope standing transport is undefined on R_H , residual metric successor support is empty,*

$$\text{Succ}_{D_0}^{\text{res}}(R_H, \text{Metric}) = \emptyset,$$

no independent residue-level re-governance is declared before exhaustion, endpoint routes are closed, and admitted nullity is unavailable. Then

$$\text{Status}(R_H) = B_A^*, \quad \text{ANF}(B_A^*),$$

and

$$\text{AENF}(B_A^*) = \text{ANF}(B_A^*) \wedge \neg \text{NullAdm}(B_A^*) \wedge \text{EndpointRouteClosure}(B_A^*).$$

No ordinary GR interior, classical singularity, remnant, archive, destruction or transfer event, energy sink, bounce, wormhole, inherited-law transition, admitted null regime, or mute-nullity shield remains as a same-scope terminal status. The result is universal over the certified post-subtraction target class: once the declared exterior quotient and all genuine certificates leave a nonempty unresolved residue with empty residual metric support, no third old-scope terminal status remains.

Proof. The derivative horizon cannot re-license standing or function as an AMetric reset. Exterior witnesses have already removed every burden they genuinely re-fix. For the nonempty remainder, standing transport is undefined and residual metric comparison has no successor support. The Endpoint Coordinate Population Lemma fixes every coordinate of the endpoint profile; False Endpoint Exhaustion removes the metric, dynamical, carrier, transport, and inherited-law rivals; Endpoint Null-Route Closure removes admitted and mute vacuity as terminal competitors. The only surviving same-scope endpoint status is B_A^* (Maley, 2026h,k,m). $\square$

Avoidance routes are target changes, not alternative endings

A certified exterior witness removes its burden before R_H is formed. Surviving residual metric support means maximum governance-depth has not been instantiated. A holographic, island,

tunneling, bounce, wormhole, or quantum-gravity bridge supplies a certificate, new operator, or richer governance scope. A non-sink quotient resolves or retypes the target. These routes may be physically substantive, but each resolves the residue, denies certification, or changes the theorem object. None supplies a third old-scope ending for the certified unresolved residue.

The boundary is not the geometric horizon. The horizon is a derivative metric/causal structure inside the licensed Einstein projection and initiates the reconstruction burden. $B_A^\star$ is fixed only when residual metric support for the unresolved residue is empty.

5.7 Absolute no-crossing

A crossing map from parent to successor would have to preserve target identity, standing, quotient, and licensed use. It would therefore be an admissible transport operator. No such operator is defined at an AMetric boundary.

Theorem 5.10 (Absolute No-Crossing, imported). *No old-scope standing-bearing structure crosses $B_A^\star$ into a post-boundary domain as inherited content. This includes target identity, standing, carrier packages, encodings, metricity, time, causality, matter and field identity, source and energy ledgers, laws, constants, symmetries, records, memories, Hilbert-space labels, branch indices, and global-state coordinates.*

A richer third-domain theory may compare, correlate, or dynamically relate structures it interprets as D_0 and D_1 . In that case the third domain is the active governance scope. The absolute-separation theorem object has changed.

Proposition 5.11 (Non-transfer is not destruction). *Parent-domain conservation and first-law ledgers remain valid inside their declared scopes, but they contain no map into D_1 :*

$$\text{Conserve}_{D_0}(e) \not\Rightarrow \text{Transfer}_{D_0 \rightarrow D_1}(e).$$

The absence of transfer implies neither creation, destruction, nor nonconservation, because those predicates require a common temporal and quantity-identification ledger.

6 AMetric normal-form homology: origin and endpoint

6.1 Role-profile equality

The proof direction is fixed: ANF and $\mathbf{a}_{\text{ANF}}$ are available before either boundary token is compared. The endpoint-side proof populates the coordinates of $\mathcal{P}_{\text{ANF}}(B_A^\star)$; it does not allow $B_A^\star$ to redefine the vector fixed at B_A^0 . The word *new* is relational to the parent governance architecture, not a new boundary species.

Lemma 6.1 (Origin-Endpoint Token Distinction).

$$B_A^\star \neq B_A^0.$$

Proof. B_A^0 is evaluated relative to admission of the parent standing-bearing interior. $B_A^\star$ is exposed only after that admission, the licensed physical interior, the Einstein projection, horizon-burden formation, exterior-witness subtraction, residual-coherence classification, and metric-support exhaustion. Identifying the tokens would require erasing the intervening standing-bearing lineage, carrying it through the boundary, or interpreting the endpoint as a temporal return. Standing monotonicity, no-repair, and no-crossing exclude those routes. $\square$

Table 6: Independent origin-side and endpoint-side derivation of the AMetric role profile.

Profile coordinate	Origin profile B_A^0	Endpoint profile B_A^*
M: metric/comparison	No metric, scale, rank, or comparison rule has primitive boundary authority.	Old-scope residual metric and comparison support is exhausted.
T: temporal/process	No primitive first instant, clock, order, or boundary process is licensed.	No old-scope temporal continuation, endpoint event, or process remains in domain.
C: causal production	No pre-admission causal relation is licensed.	No parent-to-successor causal production is defined.
O: operator domain	No admissible boundary operator exists.	No old-scope operator continues through the endpoint.
S: selector/parameter	No probability, rank, index, threshold, or canonical representative has boundary authority.	No endpoint selector, branch index, probability rule, or preferred successor law is available.
X: standing transport	No prior standing, law, ledger, or reference is inherited into the admitted interior.	No parent standing, law, ledger, or reference is inherited by a successor.
V: valuation	Standing is bivalent on the evaluated fragment; undefinedness is non-membership.	The same bivalent status holds; residual coherence does not create partial standing.
R: repair	No failed act is repaired at admission.	No singularity, bounce, remnant, archive, or reconstruction repairs the residue as the same lineage.
G: target gating	The admitted interior requires its own gate.	Any positive successor requires its own gate.
A: self-authorization	No physical or formal structure authorizes its own admission.	No exhausted metric, equation, law, or ledger re-licenses its own continuation.
Endpoint null-route closure	Mute nullity is pre-admissible, not a state.	Admitted nullity is contradictory; mute nullity is pre-admissibility.
Positive content	No physical content is fixed by the boundary alone.	No successor content or population is established by the parent theorem.

Theorem 6.2 (Origin-Endpoint Normal-Form Homology).

$$B_A^* \neq B_A^0, \quad \mathcal{P}_{\text{ANF}}(B_A^*) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}, \quad B_A^* \cong_{\text{ANF}} B_A^0.$$

Proof. The origin-side boundary theorem gives $\mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}$. The endpoint-side boundary-exposure theorem and its clause-level derivation give $\mathcal{P}_{\text{ANF}}(B_A^*) = \mathbf{a}_{\text{ANF}}$ under the endpoint antecedents. Equality of the independently populated vectors establishes $B_A^* \cong_{\text{ANF}} B_A^0$. The token-distinction lemma establishes $B_A^* \neq B_A^0$. $\square$

Proposition 6.3 (Nontriviality of the Homology Result). *The result is not a shared-name inference. The origin vector is fixed by pre-admission closure; the endpoint vector is fixed only after a standing-bearing interior, licensed Einstein projection, horizon isolation, exterior witness subtraction, residual-coherence classification, metric-support exhaustion, false-endpoint exclusion, and null-route closure. A same-scope counterexample must preserve that chain while producing a standing-relevant endpoint coordinate different from $\mathbf{a}_{\text{ANF}}$.*

Theorem 6.4 (Certified Black-Hole Instantiation of AMetric Normal-Form Homology). *Let $\mathfrak{H}_A^{\text{BH}}$ be instantiated by a fixed admissibility-bearing domain with an origin-side AMetric boundary, a standing-positive realized physical interior, a faithful Einstein projection, a certified black-hole reconstruction target, a nonempty unresolved residue, and maximum governance-depth. Then the Black-Hole AMetric Endpoint Exactness Theorem fixes*

$$\text{Status}(R_H) = B_A^*,$$

and the boundary profiles are token-distinct with equal complete AMetric role profiles:

$$B_A^* \neq B_A^0, \quad \mathcal{P}_{\text{ANF}}(B_A^*) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}, \quad B_A^* \cong_{\text{ANF}} B_A^0.$$

This conclusion is universal over the certified black-hole target class.

Proof. Origin: the admissibility-structure and AMetric theorems fix $\text{ANF}(B_A^0)$. Interior: the kernel and realized-physical-interior results license initial data, clocks, records, sources, and continuation only downstream. Metric projection: the metricity and Einstein theorems license the governed metric/source package. Residue: exterior witnesses remove only the burdens they genuinely re-fix. Endpoint: the target-class exactness theorem fixes $\text{Status}(R_H) = B_A^*$ and $\mathcal{P}_{\text{ANF}}(B_A^*) = \mathbf{a}_{\text{ANF}}$. The origin theorem gives $\mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}$, the token-distinction lemma separates the boundary positions, and equality of the vectors yields the homology. No crossing map is used. $\square$

Strong form

The certified black-hole instantiation is not merely the observation that metric language is absent at both sides. A nonmetric regime can retain probability, order, process, operatorhood, a selector, shared Hilbert space, or transport. Such a regime does not have profile $\mathbf{a}_{\text{ANF}}$. Homology holds because the complete independently derived role profiles are equal.

6.2 No physical cyclicity

Theorem 6.5 (No Physical Cyclicity). *AMetric normal-form homology does not entail a temporal path from B_A^* to B_A^0 , causal production of D_1 , common geometry, transport of information/energy/law/identity, cyclic cosmology, or populated successor existence.*

Proof. Each implication requires a role excluded by ANF. Temporal return needs order and evolution; causation needs witnessed dependence across common order; common geometry needs a spanning metric; inheritance needs a transport map; cyclic cosmology needs repeated states in one dynamical domain; populated-existence needs a positive instantiation theorem. Boundary-profile equivalence supplies none. $\square$

Corollary 6.6 (No cyclic-cosmology inference). *The theorem does not establish bounce cosmology, conformal cyclic cosmology, black-hole cosmogenesis, or a universe-production law. Such models may exist in richer scopes, but their dynamics are additional structure (Farhi et al., 1990; Penrose, 2010).*

Lemma 6.7 (No AMetric Boundary Fusion). *Let $B_A^{\star 1}$ and $B_A^{\star 2}$ be New AMetric Boundary tokens exposed by unresolved residues associated with black-hole merger components. Even if the exterior metric regime admits a single post-merger exterior quotient, there is no same-scope AMetric operation by which $B_A^{\star 1}$ and $B_A^{\star 2}$ fuse, collide, touch, join, transport, or compose. Fusion would require a common metric, temporal process, causal relation, contact relation, operator domain, or cross-boundary standing map, all excluded by AMetric boundary discipline. The admissible classification is token-distinct coexistence under an exterior classification ledger. If a later merged target exposes its own endpoint token $B_A^{\star(12)}$, that token is not the fusion of $B_A^{\star 1}$ and $B_A^{\star 2}$; it is a new endpoint classification relative to the post-merger exterior quotient.*

Proof. An exterior merger is a licensed metric/source process inside the parent exterior regime. AMetric endpoint tokens are not metric regions. To say that two such tokens touch or fuse would require a shared metric or adjacency relation; to say that one produces or joins the other would require a temporal or causal process; to say that a common endpoint operation combines them would require an operator domain; and to say that standing is shared across them would require a cross-boundary transport map. These are exactly the boundary authorities excluded by ANF. A post-merger exterior ledger may co-classify the endpoint tokens, but co-classification is bookkeeping in the exterior register, not fusion at the boundary. Therefore AMetric boundary tokens coexist without crossing. $\square$

Classification 6.8 (Homology-route normal form). Any proposed connection among $B_A^0, B_A^\star, D_0$, and a possible D_1 is exactly one of:

1. external normal-form homology, proved here;
2. independent post-boundary admission with fresh governance and no inheritance;
3. a larger-scope coupled theory $D_\star$ containing common amplitudes, Hilbert sectors, topology, laws, or ledgers; or
4. illicit crossing: same-scope transport, causation, storage, law inheritance, or cyclicity without new governance.

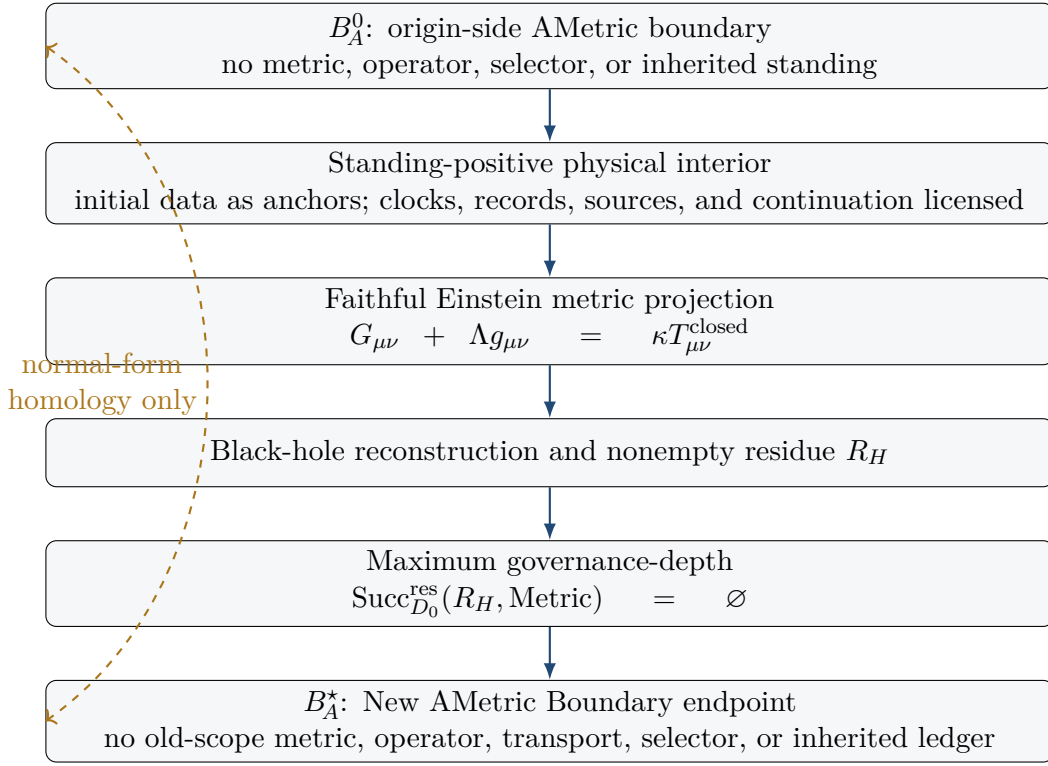


Figure 3: AMetric normal-form homology. The vertical arrows are dependency steps in the parent theorem. The dashed relation is external normal-form comparison, not a transition, worldline, or inherited map.

6.3 Existence without inheritance

Definition 6.9 (Post-AMetric domain). A post-AMetric domain D_1 is an independently governed domain admitted downstream of B_A^* . The word *post* denotes theorem dependence, not temporal succession.

Theorem 6.10 (Existence Without Inheritance). *If a non-degenerate D_1 has standing-bearing existence-status, then it instantiates its own kernel*

$$\mathcal{K}_1 = (\text{Adm}_1, \text{Stand}_1, \text{Ref}_1, \text{Irr}_1)$$

and does not inherit the parent kernel, metric, laws, energy ledger, information package, clock, or history:

$$\text{Adm}_1(D_1) = 1, \quad \text{Inherit}_{D_0 \rightarrow D_1} = 0.$$

Proof. Non-degenerate domainhood requires target fixation, admissibility, standing, reference, irreversibility, quotient stability, and continuation. Kernel necessity forces these roles in D_1 . Absolute no-crossing excludes their inheritance from D_0 . $\square$

Theorem 6.11 (No Parent-Scope Cosmography). *The parent analysis may derive endpoint status and origin-profile homology,*

$$\text{Status}(\text{GDepth}^*) = B_A^*, \quad B_A^* \cong_{\text{ANF}} B_A^0,$$

but not the contents or populated-existence of D_1 .

Proof. Endpoint status is derived from exhaustion of parent roles. A claim about successor contents requires a target-fixing witness or comparison across the boundary, excluded by no-crossing. A third-domain theory may make such claims under its own governance, but they are not parent-scope deductions. $\square$

7 Downstream application: bivalent closure for black-hole reconstruction burdens

7.1 Why this result is a corollary

The AMetric homology theorem is already complete before the following target-class result is stated. Bivalence does not generate ANF; it determines the standing dispositions available to declared black-hole reconstruction burdens inside a domain already governed by the kernel and the AMetric invariant.

Definition 7.1 (Genuine AASC-target black hole). A genuine AASC-target black hole is a downstream role package

$$\mathcal{B} = (D_0, \Pi_g, H_{\text{BH}}, \mathcal{B}_H, \mathcal{A}_{\text{wit}}, \mathcal{Q}_{\text{ext}}, \text{Cont}, \mathcal{E}) \in \mathfrak{B}_{\text{AASC}},$$

where Π_g is a licensed Einstein metric/source projection, H_{BH} is a derivative horizon-isolating role, $\mathcal{B}_H$ is the declared parent reference-standing burden, $\mathcal{A}_{\text{wit}}$ and $\mathcal{Q}_{\text{ext}}$ fix the exterior witness and quotient structures, Cont is the licensed continuation relation, and $\mathcal{E}$ contains the declared source, thermodynamic, calibration, and record ledgers. The target definition does not include an endpoint conclusion.

For $b \in \mathcal{B}_H$, define

$$\text{Res}_{D_0}(b) = \begin{cases} 1, & \text{a certified exterior witness relation re-fixes } b, \\ 0, & \text{no such relation re-fixes } b. \end{cases}$$

Let

$$\mathcal{W}_{\text{ext}}(\mathcal{B}_H) = \{b : \text{Res}_{D_0}(b) = 1\}, \quad R_H = \{b : \text{Res}_{D_0}(b) = 0\}.$$

Theorem 7.2 (Exterior Witness Partition). *For every genuine AASC-target black hole,*

$$\mathcal{B}_H = \mathcal{W}_{\text{ext}}(\mathcal{B}_H) \sqcup R_H.$$

The partition has physical force because no third same-scope old-standing disposition exists between certified exterior re-fixation and unresolved burden.

Proof. The set partition follows from the bivalent witness predicate. Suppose a third standing-relevant disposition existed. If it altered parent target identity or later use, it would require a new classifier, selector, carrier, probability, ranking, or continuation rule, contradicting AMetric no-selector/no-carrier closure or changing scope. If it altered neither classification nor continuation, it would be skin or bookkeeping. Therefore the only same-scope dispositions are certified re-fixation and unresolved burden. $\square$

Theorem 7.3 (No Partial Standing for the Residue). *For every $b \in R_H$, old-scope standing transport is undefined rather than weakly positive, probabilistically positive, conserved, latent, hidden behind the horizon, or repairable. Role-indexed residual coherence may remain, but it is not old-scope standing.*

Proof. Standing is bivalent on the evaluated fragment, while undefinedness is non-membership in the transport operator’s domain. Probabilities, entropy, reconstruction fidelity, residual metric organization, and source ledgers are downstream tensor or performance data. None supplies a third gate value. Same-lineage repair is excluded by irreversibility, and a new witness relation or larger scope is a new construction rather than retroactive positive standing. $\square$

Corollary 7.4 (Bivalent Governance Closure for Genuine Black Holes). *Every genuine AASC-target black hole has exactly one of two burden-level outcomes:*

1. *full standing-bearing exterior witness resolution, $R_H = \emptyset$; or*
2. *a nonempty unresolved residue whose old-scope standing transport is undefined and whose retained organization is at most residual coherence.*

Within the second branch, role-indexed residual coherence governs while residual metric support remains. At maximum governance-depth,

$$\text{Succ}_{D_0}^{\text{res}}(R_H, \text{Metric}) = \emptyset,$$

the Black-Hole AMetric Endpoint Exactness Theorem fixes $\text{Status}(R_H) = B_A^$. No third old-scope terminal status exists.*

Remark 7.5 (Universal application, not primitive proof). The corollary is universal over the declared post-subtraction target class. It is not the main theorem because its force is already downstream of the kernel, AMetric normal form, licensed metric interior, exterior quotient typing, and exterior witness typing. It does not use astronomical classification alone as endpoint proof. It says that once the declared quotient and all genuine certificates leave a nonempty unresolved residue, no third old-standing status remains.

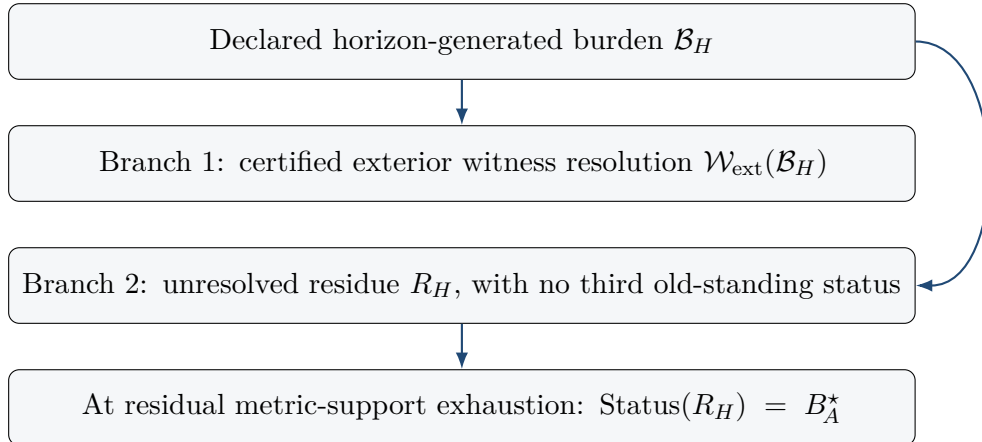


Figure 4: Bivalent black-hole burden closure as a downstream application. The partition does not define the AMetric invariant.

8 Origin-side consequences of AMetric homology

8.1 Origin-side anchoring and endpoint-side de-licensing

The homology theorem reveals a precise asymmetry that ordinary beginning/end language tends to obscure. At the origin side, metric, temporal, causal, source, and dynamical roles are not yet

licensed as boundary authority. At the endpoint side, those same old-scope roles have been licensed, used, and then exhausted for the residual target. The profiles are homologous because the same boundary-authority clauses hold, not because the histories on the two sides are the same.

Definition 8.1 (Initial-condition/endpoint duality). Initial-condition/endpoint duality is the following role correspondence:

1. origin-side initial data require an already admitted continuation envelope;
2. endpoint-side continuation data lose application when that envelope exhausts;
3. neither side permits metric, clock, source, probability, or law content to authorize the boundary at which its own standing is at issue; and
4. any positive continuation beyond either boundary requires independently fixed governance.

The duality is a dependency relation, not a time-reversal symmetry.

Table 7: Initial-condition/endpoint duality.

Role	Origin-side status	Endpoint-side status
Metric	Not available as boundary authority; may be licensed in the interior.	Previously licensed projection has no residual authority for the target.
Clock and temporal order	No primitive first instant or clock.	No old-scope final instant, duration, or transition operator.
Source/energy ledger	No primitive source quantity or conservation ledger.	Parent ledger has no cross-boundary domain of application.
Initial/final data	First interior anchor after admission.	Last metric datum remains internal to the old projection, not a boundary-crossing seed.
Probability/typicality	Cannot select an admissible start.	Cannot select a successor class or law package.
Physical law	Licensed only inside a governed domain.	Cannot be inherited into a new domain without an admissible map.
Record identity	Requires admitted carrier and witness structure.	Cannot be preserved across the boundary by bare formal similarity.
Positive domain	Requires fresh gate and closure.	Any successor requires fresh gate and closure.

Theorem 8.2 (Initial-Condition/Endpoint Duality Theorem). *Within the fixed AASC comparison class, origin-side physical initialization and endpoint-side physical termination are governed by the same no-backflow rule: no interior structure may authorize the boundary conditions of its own admissibility, and no exhausted interior structure may authorize its own cross-boundary continuation.*

Proof. At the origin, using a metric, state, probability distribution, source ledger, or clock to authorize admission is boundary backflow because the role presupposes admission. At the endpoint, using the

Einstein equation, a curvature divergence, a conservation law, a global state, or an information ledger to re-license continuation after their role-support has exhausted is endpoint backflow. Both moves make a downstream structure govern the boundary that fixes its domain of application. The AMetric no-selector, operator-undefinedness, no-transport, and independent-gating clauses exclude both. $\square$

8.2 Cauchy development and its exact limit

In standard general relativity, admissible initial data on a hypersurface satisfy the constraint equations and determine a maximal globally hyperbolic development up to diffeomorphism under the appropriate hypotheses (Choquet-Bruhat and Geroch, 1969; Wald, 1984). This is a powerful conditional theorem. It does not select the data, privilege a hypersurface independently of a model, guarantee global hyperbolicity in every spacetime, or provide a transport map outside the metric category.

AASC therefore distinguishes three statements:

1. *initial-data admissibility*: the data and hypersurface are well typed in a declared metric/source domain;
2. *metric development*: the field equations determine continuation inside that domain; and
3. *boundary authority*: the metric development remains licensed for the target under consideration.

The first two do not entail the third at a New AMetric Boundary.

Proposition 8.3 (Cauchy Development Does Not Cross an AMetric Boundary). *A maximal metric development, even when mathematically well defined, does not supply an inherited continuation map across an endpoint satisfying ANF. It remains a theorem object of the metric domain in which its hypersurface, constraints, fields, equivalence relation, and development operator are defined.*

Proof. A Cauchy development presupposes a manifold, differentiability class, metric/connection, field equations, constraint surface, and uniqueness relation up to diffeomorphism. Each is post-admission structure. Crossing an AMetric boundary would require these structures to remain the same target and to govern a newly admitted domain. Non-transmissivity and independent target gating exclude that inference. A larger theory may embed both structures, but then the larger theory is the active scope. $\square$

Remark 8.4. Mathematical continuation of a classical representative toward a singular boundary remains a metric-domain result. Mere formal writability does not certify continued physical application after the target's residual metric support has exhausted.

8.3 No inheritance of a Past Hypothesis

A low-entropy boundary condition may be specified in D_0 and may help orient thermodynamic, record, radiative, and psychological arrows inside that domain. It does not cross B_A^* as an inherited low-entropy condition for D_1 .

Theorem 8.5 (No Past-Hypothesis Inheritance). *Let H_{PH} be a Past-Hypothesis-style anchor in D_0 . If D_1 is independently admitted beyond B_A^* , no parent-scope theorem establishes H_{PH} , its macrostate partition, measure, entropy functional, or arrow orientation as inherited structure in D_1 .*

Proof. The anchor is defined relative to a parent state space, coarse-graining, measure, clock/record order, and law package. Inheritance requires an admissible identity map preserving those objects across the boundary. The no-crossing theorem excludes the map. A formally similar low-entropy anchor in D_1 would be independently licensed or externally compared, not inherited. $\square$

8.4 No inheritance of an inflationary mechanism

The same result applies to inflationary starts. A scalar field, potential, vacuum choice, background branch, and reheating channel may form a declared realization of early transport closure in a cosmological model. None is transmitted through a New AMetric Boundary as a seed for a successor cosmology.

Corollary 8.6 (No Inflationary Seed Transfer). *No parent scalar field value, inflaton potential, vacuum state, e -fold count, reheating channel, or perturbation phase is inherited by a post-AMetric domain merely because the parent endpoint is homologous to an origin condition.*

Accordingly, an “origin-condition analogy” supplies no seeding inference. The theorem establishes boundary normal-form homology only. A model containing a common scalar, common quantum state, shared wave functional, tunneling amplitude, or common path integral declares a larger scope and no longer instantiates absolute no-inheritance normal-form homology.

8.5 Standing monotonicity closes the reversal route

Standing monotonicity blocks a different overreading: that the endpoint might reverse the parent’s arrow and thereby return to the beginning. A failed standing-bearing prefix is not repaired by later continuation; old lineage cannot be re-entered as though the failure never occurred (Maley, 2026j,o).

Theorem 8.7 (No Homology by Reversal). *AMetric normal-form homology cannot be realized by reversing a parent trajectory, entropy gradient, causal order, clock sequence, or retained-record order. Any such reversal either remains an interior process before endpoint closure or presupposes precisely the old-scope ordering and operator roles unavailable at B_A^* .*

Proof. A reversal is an operation defined on an ordered trajectory. At B_A^* , no parent temporal or causal operator remains in domain. Before B_A^* , a reversal is an interior dynamical proposal and does not instantiate the boundary normal form. Moreover, standing failure in a retained lineage is irreversible; reversal cannot convert the failed lineage into its earlier standing-positive state as the same act. Thus no reversal route realizes normal-form homology. $\square$

9 Interfaces with standard physics and cosmology

9.1 Classical general relativity

The relation to general relativity has two stages. First, AASC licenses a faithful Lorentzian projection and, under the minimal first-response same-scope predicate, derives Einstein form. Second, the endpoint theorem bounds the target-relative authority of that projection. Classical solutions, causal diagrams, singularity theorems, and black-hole thermodynamics remain valid inside their declared metric scopes (Bardeen et al., 1973; Hawking and Ellis, 1973; Penrose, 1965; Wald, 1984).

The relevant scope claim is:

A metric theorem cannot classify a target after the target no longer instantiates the metric role required by the theorem’s predicates.

The metric theorem remains valid on its declared domain; the endpoint target no longer lies within that physical domain of application.

Proposition 9.1 (GR Licensing and Boundedness). *AASC preserves the Einstein projection wherever metric, source, quotient, and continuation roles retain standing or admitted residual support. It denies only the projection’s ability to self-certify its own application after that support has exhausted.*

9.2 Singularity theorems and endpoint exposure

Classical singularity theorems establish geodesic incompleteness or related metric-internal conclusions under stated hypotheses. They do not identify a physical point called “the singularity” and do not by themselves classify a nonmetric governance boundary. The homology theorem therefore preserves the singularity as the metric projection’s internal limit diagnosis while placing $B_A^\star$ at a different logical level.

Classification 9.2 (Three levels of black-hole limit language). 1. **Equation-level failure:** curvature divergence, inextendibility, anomaly, high-curvature correction, or source incompleteness inside a metric model.

2. **Projection-level exhaustion:** the licensed metric/source projection lacks residual support for the unresolved target.

3. **Boundary-level closure:** no old-scope metric, causal, temporal, operator, source-ledger, or transport authority remains, and admitted nullity is unavailable.

Only the third level has New AMetric Boundary status.

9.3 Hawking radiation, Page curves, holography, and islands

Exterior radiation and reconstruction programs bear directly on the residue antecedent. They may reduce R_H , eliminate it, or prevent maximum governance-depth from becoming the relevant target. They do not automatically refute the endpoint theorem.

Table 8: Exterior structures and their homology-theorem roles.

Structure	Admissible role	Insufficient promotion
Hawking flux (Hawking, 1975)	Parent exterior energy/thermodynamic ledger; possible witness carrier if paired with a map.	Flux alone is not reference-standing reconstruction and not a crossing channel.
Page curve (Marolf and Maxfield, 2021; Page, 1993)	Entropy/ledger profile constraining exterior models.	Turnover alone is not a target-fixing witness.
Boundary algebra	Candidate carrier for reconstruction.	Algebra without a quotient-preserving target map is not a witness.
Holographic reconstruction	Possible standing-bearing exterior witness when target, carrier, quotient, and map are fixed.	“Encoded on the boundary” is skin without the map.
Island/QES prescription (Almheiri, Engelhardt, et al., 2019; Almheiri, Mahajan, et al., 2020; Penington, 2020)	Entropy ledger or witness relation, depending on reconstruction content.	An entropy formula does not itself transport parent content across $B_A^\star$.
Global state or ensemble	Larger-scope comparator or coupled theorem object.	A shared label does not become an internal inheritance relation.

Theorem 9.3 (Exterior Resolution Precedes Endpoint Classification). *Every genuine exterior witness relation resolves its target burden before R_H is formed. It does not cross $B_A^\star$ and does not repair the remaining residue.*

Proof. The witness lands in the parent exterior or declared boundary algebra and re-fixes a parent reference class. That burden is therefore removed from R_H . The relation supplies no operator on the remaining horizon-isolated carrier and no inherited map into a successor domain. The theorem object is the unresolved remainder after this subtraction. $\square$

9.4 Baby universes, wormholes, and third-domain theories

Baby-universe and wormhole programs often contain common Hilbert spaces, path integrals, amplitudes, alpha parameters, topological relations, or superselection sectors (Farhi et al., 1990; Marolf and Maxfield, 2020, 2021). Such structures are not prohibited. They are classified as a larger governed scope D_* .

Definition 9.4 (Third-domain embedding). A third-domain embedding is a governed theory D_* containing representations of D_0 and D_1 , together with comparison or transition structure internal to D_* .

Proposition 9.5 (Third-Domain Scope Change). *A theory that supplies a common Hilbert space, path integral, topology, causal graph, energy ledger, probability distribution, or transition amplitude for parent and successor has changed the target from absolute AMetric separation to a coupled D_* -scope theory.*

This is the required scope typing. The coupled theory must then establish its own admissibility-bearing regime and show whether what appeared to be $B_A^\star$ remains AMetric relative to that larger scope. If an operator crosses it in D_* , it was not AMetric relative to that operator.

9.5 Bounce and cyclic cosmologies

Bounce and cyclic models supply a before/after relation, state space, matching law, and usually a common metric or quantum dynamics. Their bridge structure places them in a larger continuous scope, so they do not instantiate the no-crossing homology theorem as stated.

Proposition 9.6 (Cyclic-Model Classification). *A cyclic, ekpyrotic, bouncing, conformal-cyclic, or tunneling cosmology is either:*

1. *an interior dynamical model whose transition occurs before AMetric closure;*
2. *a larger-scope theory in which the alleged boundary is not AMetric relative to the bridge dynamics;*
or
3. *a target-inconsistent transition claim if it retains the absolute no-crossing target while asserting a bridge dynamics.*

9.6 Quantum-gravity response routes

A quantum-gravity theory can respond to the homology theorem in several exact ways:

1. supply exterior witness relations that make $R_H = \emptyset$;
2. replace or re-govern the carrier before residual metric support exhausts;
3. show that residual metric support never becomes empty;

4. define a richer scope in which a nonmetric but operator-bearing endpoint exists;
5. provide a common parent-successor governance domain;
6. independently admit a post-boundary domain without inheritance; or
7. break an imported AASC theorem or the endpoint inference.

What it cannot do while preserving the target is silently retain old information, energy, law, or identity across an endpoint whose defining profile excludes the needed map.

Quantum-gravity interface test

A proposed completion should state: Does it eliminate the residue, prevent maximum governance-depth, add a new carrier, add an operator, enlarge the comparison class, provide an exterior witness, or introduce independent post-boundary governance? Which old theorem antecedent is denied? Which new anchor or tensor is declared? What is the target-preserving map, if any?

10 Operational verification protocols

10.1 Origin-side audit

The origin-side verification determines whether a proposed initial condition has physical standing or performs illicit boundary authority.

Origin-side Initial-Condition Test

For a proposed initial condition I :

1. Fix the theorem object D and physical use U .
2. State whether I is a boundary anchor, tensor datum, record certificate, or presentation skin.
3. Declare the clock/order by which I is called initial.
4. Declare metric, state-space, quotient, source, gauge, record, and continuation ledgers.
5. Show invariance under lawful redescription.
6. State the failure and domain boundaries.
7. Identify every selector: probability, typicality, simplicity, regularity, symmetry, vacuum choice, branch choice, or measure.
8. If the selector is not declared as anchor/tensor content, classify hidden selection.
9. If missing structure is added after the desired history is known, classify repair.
10. If the structure changes the theorem object, classify scope enrichment.

10.2 Endpoint audit

The endpoint verification determines whether a proposed black-hole endpoint has the roles required by its own description.

Endpoint Test for an Unresolved Residue

1. Fix the parent domain, Einstein projection, horizon burden, and exterior witness algebra.
2. Remove every burden genuinely re-fixed by an exterior witness relation.
3. Confirm whether $R_H \neq \emptyset$.
4. Confirm that old-scope standing transport is undefined for every retained burden.
5. Distinguish residual coherence from standing.
6. Determine whether residual metric successor support is empty.
7. Test each endpoint proposal for required roles: metric, affine, temporal, causal, operator, carrier, ledger, map, or selector.
8. Classify declared richer structure as scope change rather than hidden repair.
9. Test nullity: mute, non-null governance, or contradiction.
10. Verify ANF clause by clause before reporting New AMetric Boundary status.

10.3 Normal-form homology claim audit

The normal-form homology verification prevents theorem-level homology from being laundered into cosmology.

AMetric Homology Claim Test

A homology claim is reportable only if:

1. both boundary profiles independently satisfy ANF;
2. the comparison is role-profile equivalence, not token identity;
3. no physical arrow is drawn from $B_A^\star$ to B_A^0 ;
4. no common time, metric, law, energy ledger, state, information package, or probability measure is inherited;
5. no populated successor is inferred from endpoint status;
6. any larger-scope relation is explicitly assigned to D_* ;
7. cyclic, bounce, tunneling, and baby-universe language is marked comparator or richer-scope language; and
8. the conclusion is limited to origin-endpoint normal-form homology.

10.4 Counterexample normal form

A substantive same-scope counterexample must satisfy all of the following:

1. preserve the AASC kernel and fixed comparison class;
2. admit primitive physical initial data that authorize their own clock, metric, source, record, and continuation structures without selector or backflow; or preserve a black-hole residue with empty

residual metric support while supplying a coherent non-AMetric endpoint;

3. avoid adding new anchors, tensors, operators, boundary channels, Hilbert spaces, measures, or target gates;
4. avoid representative privilege, repair, deferred admission, second classifier, carrier laundering, and scope transport;
5. provide a physically standing-bearing distinction that alters neither evaluated classification nor retained continuation; and
6. preserve cross-boundary identity without an admissible transport map or external common scope.

The conjunction is the refutation burden. A proposal satisfying only some items is a different target, a physical realization in a richer scope, or a restatement of one of the closed routes.

11 Main theorem package

11.1 Theorem ladder in dependency order

The argument is separated into the following ordered results.

1. **No Downstream Authorization:** metric, causal, source, horizon, and endpoint structures cannot authorize the kernel or AMetric invariant.
2. **Origin-Side Boundary Theorem:** $\text{ANF}(B_A^0)$.
3. **Initial-Condition Standing and Firewall:** physical initial data are downstream anchors, tensors, or certificates and cannot authorize their own standing package.
4. **Licensed Physical Interior and Einstein Projection:** metric/source/clock/record structures are lawful downstream roles and carry no boundary backflow.
5. **Endpoint-Side Boundary Exposure:** a downstream regime whose old-scope role support exhausts under route closure satisfies ANF.
6. **AMetric Normal-Form Homology:** $B_A^\star \neq B_A^0$, $\mathcal{P}_{\text{ANF}}(B_A^\star) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}$, and $B_A^\star \cong_{\text{ANF}} B_A^0$.
7. **No Crossing and No Inheritance:** no old-scope carrier, law, metric, clock, information, energy, or causal relation is transported through $B_A^\star$.
8. **Target-Class Exact Black-Hole Endpoint:** a certified unresolved residue at maximum governance-depth has endpoint status $B_A^\star$, with no third old-scope terminal profile.
9. **Bivalent Black-Hole Closure Corollary:** every declared target-class burden is externally re-fixed or unresolved; no third old-standing status exists.

11.2 Homology capstone theorem

Theorem 11.1 (AMetric Origin-Endpoint Normal-Form Homology Theorem). *Assume the AASC kernel and the regime-invariant AMetric boundary theorem. Let B_A^0 be the origin-side boundary profile of an admitted physical interior. Let $B_A^\star$ be the endpoint profile of a downstream governed regime satisfying:*

- (A1) *the metric, temporal, causal, source-ledger, record, and operator structures used in the regime were licensed downstream of the kernel and ANF;*

- (A2) *those old-scope roles have no standing-bearing support for the endpoint target;*
 (A3) *no independent re-governance has been declared before endpoint classification;*
 (A4) *selector import, representative privilege, repair, inherited transport, storage, dynamical transition, and second-classifier routes are unavailable; and*
 (A5) *admitted total vacuity is unavailable as a same-type endpoint.*

Then

$$\text{ANF}(B_A^0), \quad \text{ANF}(B_A^\star),$$

$$B_A^\star \neq B_A^0, \quad \mathcal{P}_{\text{ANF}}(B_A^\star) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}, \quad B_A^\star \cong_{\text{ANF}} B_A^0.$$

No physical loop, causal production, common time, crossing map, information or energy transfer, inherited law, or populated successor follows.

Proof. First, the kernel and AMetric theorem establish $\mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}$ independently of all physical-interior content. Second, no downstream authorization prevents later metric or endpoint structures from modifying that profile. Third, under (A1)–(A4), the endpoint clause table eliminates every metric, temporal, causal, operator, selector, carrier, transport, repair, and self-relicensing route; under (A5), admitted nullity is excluded while mute nullity has no rival force. Hence $\mathcal{P}_{\text{ANF}}(B_A^\star) = \mathbf{a}_{\text{ANF}}$. The equivalence between ANF and vector equality is used only as a coordinate representation of the fixed normal form; the proof remains the independent population of the two role profiles. Fourth, the origin-endpoint token-distinction lemma establishes $B_A^\star \neq B_A^0$. Fifth, equality of the independently derived vectors proves $B_A^\star \cong_{\text{ANF}} B_A^0$. The excluded coordinates are precisely those required for cyclicity, causation, crossing, inheritance, and populated-successor conclusions. $\square$

Corollary 11.2 (Black-Hole Endpoint Exactness). *For every black-hole package instantiating $\mathfrak{H}_A^{\text{BH}}$, the unresolved residue has endpoint status $\text{Status}(R_H) = B_A^\star$. The result is exact over the certified target class; resolution before exhaustion, denial of certification, or prior richer governance changes the target rather than supplying another old-scope endpoint.*

Corollary 11.3 (Bivalent Black-Hole Governance Closure). *For every genuine AASC-target black hole, $\mathcal{B}_H = \mathcal{W}_{\text{ext}}(\mathcal{B}_H) \sqcup R_H$, with no third old-standing disposition. This corollary follows from the primitive bivalence and witness disciplines; it is not used to prove the homology theorem.*

11.3 Operational minimality

Table 9: Deletion audit for the homology capstone theorem.

Deleted role/result	Immediate loss	Resulting failure
Kernel/irreversibility	No fail-closed standing or no-repair.	Boundary and endpoint acts can be retargeted after failure.
AMetric invariant	No independently fixed normal form.	Origin and endpoint comparison becomes stipulative.
No downstream authorization	Metric or black-hole content can be laundered into boundary authority.	Dependency order reverses.

Deleted role/result	Immediate loss	Resulting failure
Initial-condition discipline	Physical data can be mistaken for primitive starts.	Origin-side application is lost.
Metric licensing and Einstein projection	No exact downstream metric/source object is fixed.	Endpoint exposure becomes an external rejection of GR.
Standing monotonicity	No retained-lineage anti-reversal discipline.	Homology can be misread as temporal return.
Endpoint route exhaustion	Non-AMetric endpoint categories remain open.	$\text{ANF}(B_A^*)$ is unproved.
Null-regime closure	Vacuity remains an apparent rival.	Endpoint normal form is incomplete.
No-crossing	Homology can be laundered into inheritance.	Cyclic and baby-universe overclaims reopen.
Black-hole residue discipline	The physical application lacks an exact target.	Exterior ledgers and slogans can be confused with witness resolution.

11.4 Counterexample conditions

A substantive refutation must proceed by one of four routes:

1. break an imported kernel, AMetric, metricity, Einstein, trajectory, residual-coherence, null, or endpoint theorem;
2. identify a misuse of an imported result;
3. show that the capstone antecedents cannot jointly instantiate; or
4. exhibit a same-scope survivor: an endpoint relation or boundary authority that preserves the comparison class while escaping evaluated classification, retained continuation, anchor–tensor–skin typing, role-profile equality, route collapse, and no-crossing.

These are structural counterexample conditions; changes of target or scope are distinct theorem objects.

12 Referee standing matrix and exact counterexample routes

Referee objections enter the proof only through the correctness routes fixed above. The matrix identifies the witness required for standing. Without that witness, the objection is an overreporting correction, target denial, scope change, or request for another theorem class rather than a same-scope refutation.

Table 10: Standing requirements for common objection classes.

Objection class	Required standing witness	Status without the witness
Black-hole failure is said to generate AMetricity.	A dependency-preserving derivation in which the endpoint establishes the kernel or ANF without presupposing them.	Boundary backflow.
The metric interior must validate the invariant.	A metric structure that supplies its own admissibility, comparison, quotient, and continuation conditions without circularity.	Dependency reversal.
The homology result is definitional.	A coordinate of either boundary profile that is assigned rather than independently derived, or a failure of profile sufficiency.	Nonstanding naming objection.
The black-hole endpoint is merely conditional or one optional ending.	A denial of target certification, certified exterior resolution, surviving residual metric support, prior richer governance, failure of a load-bearing theorem, or a same-scope terminal survivor.	Misreading of target-class theorem form.
Nullity remains an endpoint rival.	A same-type admitted-null regime or mute-nullity classifier that preserves vacuity while carrying positive endpoint force.	Refuted null route or non-governing pre-admissibility.
Residual coherence is an arbitrary name for whatever remains.	A post-break role description lacking pre-break role ancestry, retained organization, standing loss, and failure of full role licensing while still satisfying the residual-coherence predicate.	Failure to meet the defined target.
The horizon can re-license standing as a new origin.	A same-scope horizon that licenses metric, causal, and standing roles without presupposing them, adding a certificate, or declaring new governance.	Horizon-as-AMetric-reset backflow.
The same boundary returns, or the universe is cyclic.	A common time, state space, evolution operator, and identity map preserving the fixed no-crossing target.	Prohibited promotion.
The endpoint creates or guarantees a successor.	An independently admitted production or population theorem internal to the parent scope.	Unsupported positive instantiation.
Every nonmetric endpoint is being called AMetric.	A same-scope endpoint preserving probability, selector, operator, transport, repair, or self-authorization while satisfying the stated antecedents.	Different role profile or different target.

Objection class	Required standing witness	Status without the witness
A singularity or smooth horizon decides endpoint status.	A proof that metric-internal diagnosis or local smoothness preserves residual metric authority for the unresolved target.	Level conflation.
Hawking radiation, a Page curve, holography, or islands resolve the residue.	A certified target-fixing exterior witness map for each disputed burden.	Ledger or presentation only.
Unitarity, information, or energy must persist across the boundary.	A common Hilbert space, quantity-identification ledger, carrier, and admissible cross-boundary operator preserving the same scope.	Hidden transport or richer scope.
Probability or tunneling selects a successor.	A declared sample space or quantum operator whose domain spans the boundary without changing the theorem object.	Selector import or scope enrichment.
A larger theory relates parent and successor.	A demonstration that the larger theory preserves rather than replaces the fixed D_0 -scope theorem object.	Third-domain theory D_* .
Empirical inaccessibility makes the theorem empty.	A correctness witness showing failure of target certification, dependency, profile derivation, or route exhaustion.	Nonsequitur about observability.
Governance is not physical.	A physical theorem that uses no target fixation, quotient identity, operator domain, continuation rule, or failure boundary.	Category mistake.

12.1 Target-certification denial

A standing target objection identifies a failed certification condition in the declared origin-endpoint package. For the black-hole specialization this may be $R_H = \emptyset$, survival of residual metric support, a certified exterior witness for every burden, or prior independent re-governance. Such a result resolves, denies, or changes the target before endpoint classification. It is not another ending of the same certified unresolved residue, does not reverse the dependency order, and does not alter the general capstone theorem.

12.2 Imported theorem breakage

A standing dependency objection names the imported theorem, states the hypothesis instantiated here, and exhibits a counterexample to its conclusion. Breaking the kernel, AMetric comparison-class lock, bivalence, anchor–tensor–skin exhaustion, metricity licensing, Einstein-projection scope, standing monotonicity, residual-coherence theorem, endpoint closure, no-crossing, or null-regime result breaks the corresponding local step. General disagreement with AASC terminology is not such a break.

12.3 Invalid specialization or dependency misuse

A standing specialization objection shows that an imported conclusion is applied outside its declared hypotheses or made to perform work it does not license. Examples would include using the black-hole endpoint theorem to establish ANF, treating Einstein projection as universal gravity, promoting residual coherence to partial standing, or using no-crossing to deny an explicitly larger D_* -scope theory. The dependency ledgers make each such burden local and testable.

12.4 Same-scope endpoint survivor

A direct counterexample preserves $\mathfrak{H}_A^{\text{BH}}$, the fixed comparison class, the nonempty unresolved target, empty residual metric support, route closure, and null closure, yet supplies a standing-bearing endpoint coordinate different from $\mathbf{a}_{\text{ANF}}$. The survivor must not add a metric, clock, selector, probability measure, operator, carrier, common Hilbert space, law package, boundary channel, repair rule, or external domain. Producing such a coordinate refutes endpoint profile exhaustion.

12.5 Cross-boundary survivor

A standing no-crossing objection supplies a map preserving target identity, standing, quotient, law, ledger, and licensed use across B_A^* while remaining neither an operator, transport rule, storage carrier, selector, repair, independently governed target, nor relation internal to D_* . A common state label or formal isomorphism is insufficient. The burden is an admissible same-scope transport survivor.

12.6 Third black-hole burden status

A standing objection to bivalent burden closure exhibits a burden that is neither re-fixed by a certified exterior witness nor unresolved, yet remains same-scope and standing-relevant without a new classifier, probability value, latent carrier, or richer-scope relation. Performance grades, entropy values, reconstruction fidelity, and residual organization remain interior data; they do not constitute a third standing status.

Nonstanding requests

Requests for a preferred mechanism, familiar vocabulary, cyclic interpretation, successor population, softer selector, endpoint handoff from an astronomical label, or journal-conventional ontology do not engage the proof unless they instantiate one of the standing routes above. They are not answered by weakening the theorem object.

Standing refutation criterion

The theorem is refuted only by target non-instantiation that defeats the stated application, failure of a load-bearing imported theorem, invalid dependency use, inconsistency of the capstone antecedents, failure of independent coordinate derivation, or a faithful same-scope survivor. The burden is structural and exact.

Arc handoff summary

The homology theorem ends at token-distinct AMetric role-profile equality plus no-crossing and no-inheritance. What is exported forward is the completed homology handoff: $B_A^* \neq B_A^0$, equal normal-form profiles, null-route closure, absolute no-crossing, and no populated successor from

homology alone. The capstone alone may ask which counter-boundary status-class survives after those locks are imposed.

13 Conclusion

Series-position conclusion. This paper exports AMetric normal-form homology only: token distinction, role-profile equality, endpoint exactness for the certified target, no-crossing, and no inheritance. It does not select the counter-boundary survivor. The capstone paper begins where this homology result stops. The dependency order is the result's first constraint. The AASC kernel fixes admissibility, standing, reference, and irreversibility. The AMetric theorem fixes the boundary normal form and its role vector $\mathbf{a}_{\text{ANF}}$. Neither is inferred from initial data, metric geometry, Einstein dynamics, horizons, singularities, black-hole residues, or endpoint behavior. Those structures become physically usable only inside an already admitted interior.

Origin-side initial conditions are therefore not primitive inputs. A Cauchy datum, low-entropy condition, vacuum choice, scalar-field value, or cosmological boundary assumption can be indispensable to a physical model, but it enters as an anchor, quotient-level tensor, or transition certificate after the comparison class, clock, metric, source, record, and continuation packages are fixed. It does not authorize those packages.

The admitted interior may license Einsteinian metric dynamics,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}},$$

as its minimal first-response metric/source projection. This projection is preserved within its declared domain and bounded at its application limit. When a downstream regime loses the residual support required for its old-scope metric, temporal, causal, source-ledger, and operator descriptions, the projection does not generate a new boundary. The independently fixed AMetric invariant classifies the endpoint profile coordinate by coordinate.

The black-hole residue case supplies the certified physical instantiation. Exterior witness relations resolve only the burdens they genuinely re-fix. For the certified unresolved target, once old-scope standing transport is undefined, residual metric successor support is empty, no independent re-governance precedes exhaustion, and null closure holds, the endpoint is B_A^* . This is not an optional ending within AASC. Avoidance requires prior resolution, denial of certification, or a richer governance scope before exhaustion. The bivalent burden corollary remains downstream: each burden is re-fixed or unresolved, and an unresolved burden has at most residual coherence rather than partial standing.

The homology capstone is

$$B_A^* \neq B_A^0, \quad \mathcal{P}_{\text{ANF}}(B_A^*) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}, \quad B_A^* \cong_{\text{ANF}} B_A^0.$$

The boundary profiles are token-distinct and role-profile identical. The same AMetric invariant governs origin-side licensing and endpoint-side old-scope exhaustion. No physical loop, common time, causal production, crossing map, information or energy transfer, inherited law, or populated successor follows. Any positive post-boundary domain requires its own gate and governance.

The AMetric invariant is not authorized by the metric interior. The metric interior exists only downstream of it.

The theorem is closed under its stated correctness routes. A responsive challenge must deny target certification, break a load-bearing dependency, identify an invalid specialization, prove antecedent incompatibility, or exhibit a faithful same-scope survivor. Requests for another mechanism or a richer cosmology are distinct theorem objects.

A Formal dictionary

Table 11: Formal and role-theoretic dictionary.

Symbol or term	Meaning
$\mathcal{K} = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$	Minimal AASC kernel: admissibility, standing, reference, and irreversibility.
B_A^0	Origin-side AMetric boundary profile; the superscript is theorem position, not time zero.
$B_A^\star$	New AMetric Boundary endpoint profile at maximum governance-depth.
$\text{ANF}(B)$	Strict AMetric boundary normal-form predicate N1–N8.
$\text{AENF}(B)$	Endpoint normal form: $\text{ANF}(B)$ plus null-route and endpoint-route closure.
$\mathcal{P}_{\text{ANF}}(B)$	Complete boundary role-profile map for metric, temporal, causal, operator, selector, transport, valuation, repair, gating, and self-authorization status.
$\mathbf{a}_{\text{ANF}}$	Canonical AMetric role vector $(0, 0, 0, 0, 0, 0, \text{bi}, 0, \text{ind}, 0)$.
$\mathfrak{H}_A^{\text{gen}}$	General origin-endpoint theorem package: kernel, invariant, role-profile coordinates, origin token, abstract endpoint profile, route closure, null closure, and no-crossing.
$\mathfrak{H}_A^{\text{BH}}$	Certified black-hole specialization adding parent scope, licensed metric/source projection, derivative horizon, witness subtraction, unresolved residue, and residual-successor support.
$\mathfrak{I}$	Initial-data certificate package.
$\mathfrak{M}$	Licensed metric/source continuation package.
D_0	Parent standing-bearing domain containing the licensed black-hole metric projection.
D_1	Conditional independently governed post-AMetric domain.
$D_\star$	Larger third-domain theory representing or coupling parent and successor.
$\mathcal{B}_H$	Horizon-generated parent reference-standing burden.
$\mathcal{W}_{\text{ext}}(\mathcal{B}_H)$	Portion of the burden resolved by genuine exterior witness relations.
R_H	Unresolved horizon-isolated residue $\mathcal{B}_H \setminus \mathcal{W}_{\text{ext}}(\mathcal{B}_H)$.
Residual coherence	Retained role organization after standing loss; not a third standing value.
Governance-depth	Order on thinning residual successor support; not a distance or proper time.
Maximum governance-depth	Empty residual metric successor support for the target.
$B_A^\star \cong_{\text{ANF}} B_A^0$	Equality $\mathcal{P}_{\text{ANF}}(B_A^\star) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}$, not token identity or transport.

Symbol or term	Meaning
Initial-condition standing	Physical eligibility of initial data inside a declared continuation package.
Boundary backflow	Use of downstream metric, clock, probability, law, or state content to authorize its own boundary standing.
Endpoint backflow	Use of an exhausted projection or ledger to re-license its own cross-boundary continuation.
No-crossing	Absence of inherited standing-bearing transport across an AMetric boundary.
Existence-status	Structural domainhood under a domain's own governance.
Populated-existence	Concrete internal contents, laws, fields, metricity, or physical occupancy.
Standing objection	A target-preserving correctness challenge that breaks a dependency, shows misuse or antecedent inconsistency, or supplies a faithful same-scope survivor.

B Proof dependency graph

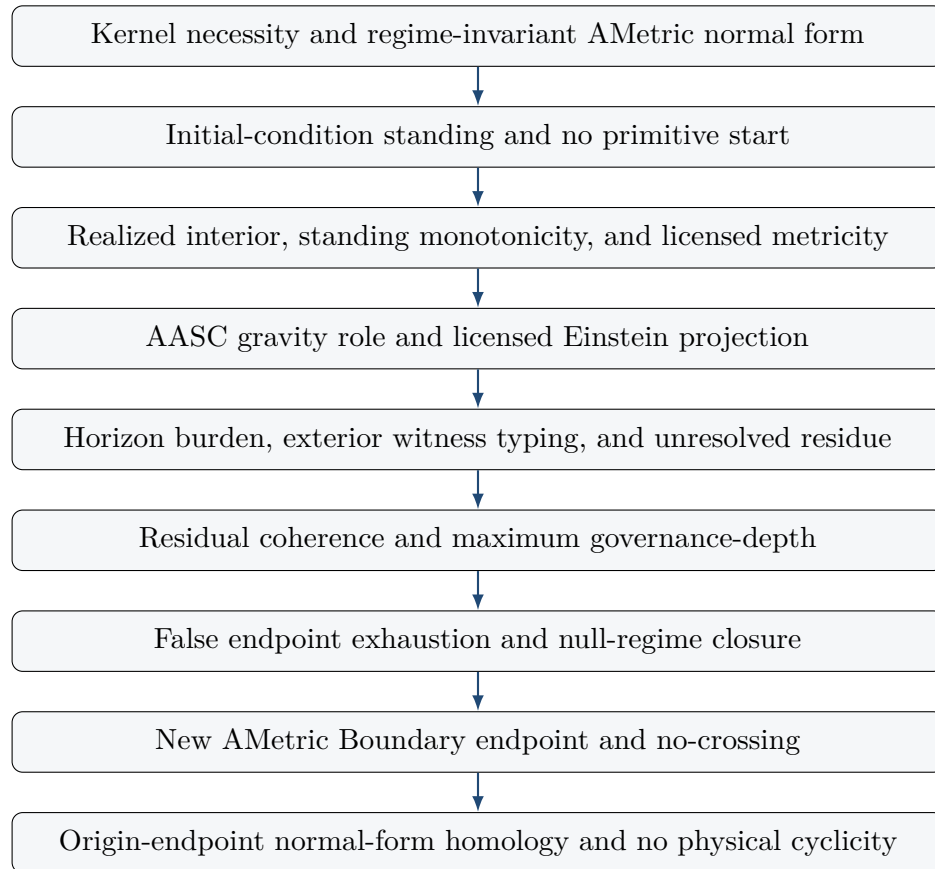


Figure 5: Dependency graph for the homology theorem. Every vertical arrow is a proof dependency, not a cross-boundary physical process.

C Claim-status ledger

Table 12: Modal and reportability status of central claims.

Claim	Status	Basis
Primitive physical initial conditions are unavailable.	Proved in target class.	Initial-condition standing theorem and AMetric closure.
Ordinary initial data are legitimate model inputs.	Preserved.	Declared anchors/tensors inside a licensed scope.
Metricity is primitive boundary structure.	Denied.	Metric licensing and no-backflow.
Einstein form is the licensed minimal first-response projection.	Imported theorem.	AASC gravity plus curvature/-source continuation exhaustion.
Standing monotonicity supplies interior direction.	Imported theorem.	Retained consequence and no-repair.
A black-hole model with nonempty post-subtraction residue and exhausted residual metric support reaches the endpoint target.	Target-class exact after antecedents.	Exterior quotient/certificate subtraction plus residual-support exhaustion.
A certified nonempty unresolved residue at maximum governance-depth has New AMetric Boundary status.	Target-class exact.	Black-Hole AMetric Endpoint Exactness Theorem.
The origin and endpoint boundary profiles are homologous.	Proved.	Independent derivation and equality of the complete role profiles.
A post-boundary domain exists and is populated.	Not proved.	Requires independent positive instantiation.
Old content, law, information, or energy crosses.	Denied.	AMetric non-transmissivity and no-crossing.
The universe is cyclic.	Not implied.	No physical-cyclicity theorem.
A larger common theory may relate both domains.	Conditionally permitted.	Scope change to D_* .

D Counterexample burden

A same-type counterexample to the capstone must construct a fixed admissibility-bearing parent domain satisfying the imported kernel and comparison class, admit legitimate initial data while showing that one primitive datum authorizes its own standing conditions, preserve a nonempty unresolved black-hole residue, preserve old-scope standing failure, preserve empty residual metric support, reject admitted nullity, and nevertheless supply a coherent non-AMetric endpoint carrying same-type physical force. It must do so without a new selector, metric, probability, operator, state space, law package, external witness, target gate, source ledger, common Hilbert space, or scope change. It must also establish cross-boundary identity without an admissible map.

Equivalently, a critic may break one of the imported theorems or identify a misuse. The dependency architecture exposes the exact location at which a rebuttal must operate.

What is not a same-type counterexample

A bounce with a matching law, a wormhole with shared geometry, a tunneling amplitude, a global wavefunction, a holographic reconstruction map, an exterior decoding algebra, a successor probability distribution, a new scalar field, or a larger conservation ledger may all define serious physical programs. Each adds standing-bearing structure and therefore changes an antecedent, resolves the residue, or enlarges the scope. None is a same-scope survivor merely by retaining the word “boundary.”

E Imported theorem interfaces and source provenance

The proof uses a dependency-ordered set of imported AASC results: kernel necessity; the admissibility-side structure theorem; the regime-invariant AMetric boundary theorem; unique fixed-domain governance; initial-condition standing; standing monotonicity; anchor–tensor–skin decomposition; metric licensing; the realized physical interior and AASC gravity role; the licensed Einstein projection; residual coherence and no horizon-as-AMetric-reset; black-hole endpoint exactness and no-crossing; and null-regime closure.

The homology capstone theorem object is not any imported result individually and is not the bivalent black-hole burden partition. It is the token-distinct equality

$$B_A^\star \neq B_A^0, \quad \mathcal{P}_{\text{ANF}}(B_A^\star) = \mathcal{P}_{\text{ANF}}(B_A^0) = \mathbf{a}_{\text{ANF}}.$$

The dependency-ordered proof fixes the origin profile, types physical initial data, licenses the downstream physical and metric/source envelope, derives the endpoint coordinates under route closure, proves token distinction, establishes role-profile equality without a crossing map, and closes cyclicity, inheritance, causal production, and populated-successor promotion.

Table 13: Imported theorem interfaces, exact local use, and prohibited overuse.

Source and archival pointer	Imported hypothesis or closure	Exact local output	Prohibited overuse
Kernel necessity	Non-degenerate identity-preserving construction forces admissibility, standing, reference, irreversibility, target identity, and no same-act repair.	Dependency-prior governance package.	Does not supply metric physics, endpoint attainment, or successor content.
Structure of Admissibility Zenodo 19198249	Equality-authoritative proposal side; arbitrary gate-bearing status interface; fixed-domain continuation.	AMetricity, bivalence, standing as reuse-stable admissibility, unique admissible interior, and no creation, transfer, amplification, or repair of standing.	Does not establish the broader AMetric export theorem or physical interiors.

Source and archival pointer	Imported hypothesis or closure	Exact local output	Prohibited overuse
AMetric Boundary Zenodo 18603144	Fixed comparison class; no selector or parameter; behavior and route exhaustion; independent gating.	Complete boundary normal form, role-vector coordinates, export discipline, and endpoint classification rule.	Does not select initial data, manufacture endpoints, or populate domains.
Unus/unique interior	No competing same-domain gates, repairs, generators, carriers, or discrete admissible interiors.	Blocks rival same-domain governance while preserving independently gated domains.	Does not imply one total domain or one interior clock, field, or solution.
Initial-condition closure and ATS	No privileged physical start; anchor–tensor–skin role exhaustion.	Types initial data as downstream anchors, tensors, certificates, or skin.	Does not deny ordinary model inputs or model-relative initial-value practice.
Standing monotonicity and trajectory theory Metricity as Licensed Description Zenodo 20338474	Retained-consequence orientation, prefix closure, and irrecoverability. Metric comparison is licensed only downstream of standing-positive structure.	Token distinction, no reversal, and no same-lineage return. Metricity as bounded interior articulation and no metric backflow.	Does not generate a global time coordinate. Does not deny metric models or license boundary self-authorization.
Realized physical interior and gravity role	Metric, clock, source, record, causal, field, and continuation roles are downstream and quotient-governed.	Standing-positive physical envelope in which a metric/source projection can be instantiated.	Does not independently fix the Einstein tensor or endpoint status.
Einstein Dynamics Zenodo 20538579	Four-dimensional Lorentzian minimal first-response same-scope class with a closed source ledger.	Licensed Einstein metric/source form used as the declared black-hole application envelope.	Does not establish universal gravity, AMetricity, or post-boundary dynamics.
Standingless Interiors Zenodo 20727246	Standing break, derivative horizon, no horizon-as-AMetric-reset, retained role ancestry, and residual-successor support.	Residual coherence, governance-depth, metric-depth failure, and re-governance classification.	Does not create partial standing, a hidden carrier, or endpoint status before support exhaustion.
New AMetric Boundary of Black Holes			

Source and archival pointer	Imported hypothesis or closure	Exact local output	Prohibited overuse
Zenodo 20773784	Nonempty unresolved residue, undefined old-scope standing transport, exhausted residual metric support, route closure, and independent gating.	Target-class exact endpoint token B_A^* , no-crossing, and no inheritance.	Does not prove every observed black hole reaches maximum governance-depth or that a successor exists.
Structural Inadmissibility of the Null Regime Zenodo 20074284	Fixed-domain admissibility-bearing evaluability and total AASC vacuity.	Admitted-nullity exclusion, mute-nullity classification as AMetric pre-admissibility, and same-type null-route closure.	Does not instantiate positive content or prove unrestricted ontology.

The project archive includes `SOURCE_MANIFEST.md`, which records exact source filenames, archival DOI pointers where available, page counts, local theorem roles, and SHA-256 hashes, together with the source PDFs and corpus-control extracts used for dependency verification. Correctness depends on the imported statements and the local inferences above, not on source familiarity or external status labels.

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